

Evolution of Computers

Learning Goals

- Early Counting Tools
- Abacus
- Pascaline Adding Machine
- Leibniz Step Reckoner
- Charles Babbage's Analytical Engine
- Lady Ada Lovelace's Programs
- Herman Hollerith's Tabulating Machine
- Computer Generations



Let's Warm Up the Brain

How Would You Count Without Numbers?

Imagine you are living thousands of years ago when there are no calculators or computers.

Situation	Object Used
Counting goats	
Counting bags of grains	
Keeping track of days	
Counting fish caught	

Hint: Pebbles, Knotted Rope, Sticks, Shells

Computers have become an important part of our everyday life. We use them to study, play games, communicate, create projects, and solve many kinds of problems. However, computers were not invented overnight. They developed gradually over thousands of years as people searched for faster and more accurate ways to count and perform calculations. In this chapter, you will explore how computers evolved, learn about the remarkable inventions that shaped their history.

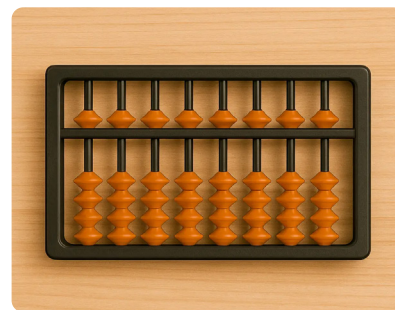
Early Counting Tools

Long before computers and calculators were invented, people still needed to count and keep records. They used simple objects found in nature, such as fingers, pebbles, shells, sticks, seeds, and bones. Some people also tied knots in ropes or made marks on stones and clay tablets. Although these methods were simple, they helped people count animals, food, money, and days. These early counting methods were the first step in the journey toward modern computers.



Abacus – The First Calculating Device

The Abacus is one of the oldest calculating devices ever invented. It was developed thousands of years ago and was widely used in countries such as China. It consists of a wooden frame with rods and colourful beads that slide from one side to the other. By moving the beads, people could perform addition, subtraction, multiplication, and division quickly and accurately. Even today, the Abacus is used in many schools to improve children's mental calculation skills and concentration.



Tech Fact

Many students who regularly practise using an Abacus become faster at solving mathematical problems in their minds without using a calculator.

Pascaline Adding Machine

In 1642, the French mathematician Blaise Pascal invented a mechanical calculating machine called the Pascaline. It was designed to help perform mathematical calculations using a set of rotating wheels and gears. The Pascaline could solve addition and subtraction problems quickly and reduced the chance of human error. It became one of the earliest successful mechanical calculators and inspired future inventors to develop better calculating machines.



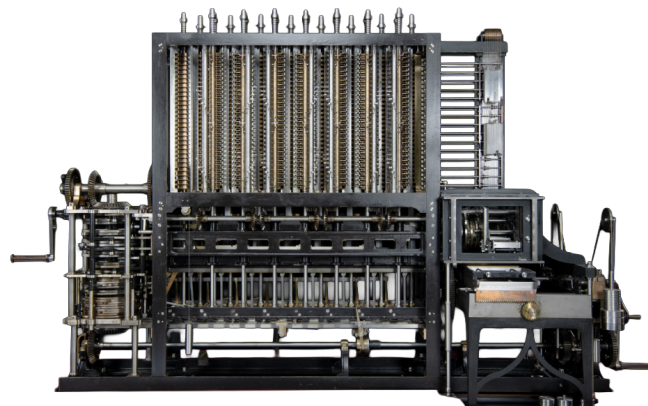
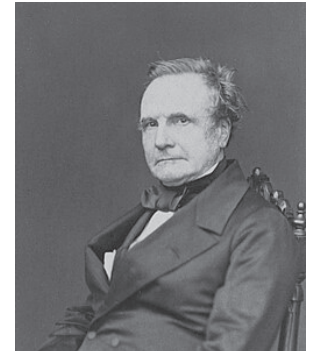
Leibniz Step Reckoner

A few years later, the German mathematician Gottfried Wilhelm Leibniz improved Pascal's invention by creating the Step Reckoner. This machine could perform all four basic arithmetic operations – addition, subtraction, multiplication, and division. It used a special gear mechanism that made calculations faster and more accurate. The Step Reckoner became an important milestone in the history of calculating machines.



Charles Babbage's Analytical Engine

The English mathematician Charles Babbage dreamed of building a machine that could perform different types of calculations automatically. He first designed the Difference Engine and later planned a more advanced machine called the Analytical Engine. This machine included features similar to those found in modern computers, such as a memory unit, a processing unit, and the ability to follow instructions. Although the machine was never fully completed during his lifetime, Charles Babbage is remembered as the Father of the Computer because of his remarkable ideas.



Lady Ada Lovelace's Programs

Lady Ada Lovelace worked closely with Charles Babbage and understood the great potential of the Analytical Engine. She wrote detailed instructions that explained how the machine could solve mathematical problems. These instructions are considered the world's first computer program. Because of her pioneering work, Ada Lovelace is known as the First Computer Programmer. Her ideas proved that computers could do much more than simple calculations.



Herman Hollerith's Tabulating Machine

In 1890, Herman Hollerith developed the Tabulating Machine to process large amounts of information quickly. The machine used punched cards to store and read data automatically. It greatly reduced the time needed to count and organise information during the population census. Hollerith's invention later led to the formation of a company that eventually became IBM (International Business Machines), one of the world's leading computer companies.



Tech Fact

Herman Hollerith's punched-card technology was used for many years before electronic computers became common.



Computer Generations

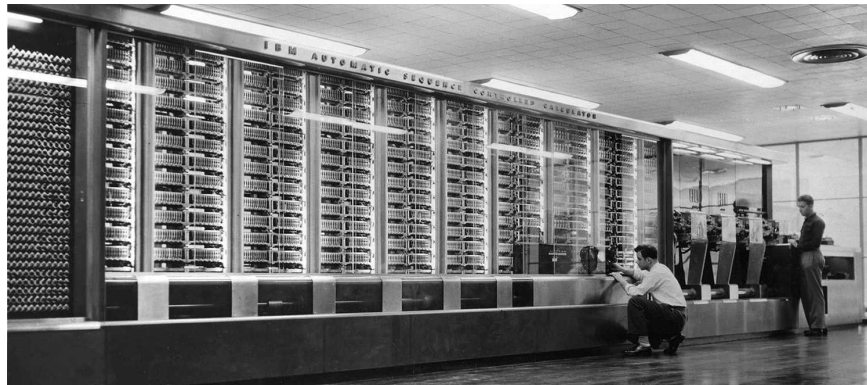
As technology improved, computers became faster, smaller, and more powerful. The development of computers is divided into five generations, each marked by important technological improvements.

First Generation (1940s)

The first generation computers were built using vacuum tubes. These computers were very large in size and occupied entire rooms.

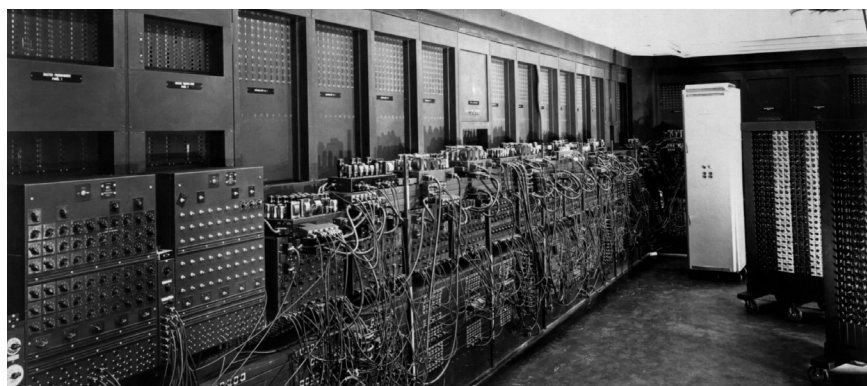
MARK-I (The First Electro-Mechanical Computer)

In 1944, Howard Aiken developed the MARK-I, one of the earliest electro-mechanical computers. It used switches, relays, and punched paper tape to perform calculations. It was more reliable than many earlier calculating machines and was mainly used for scientific calculations.



ENIAC – The First Electronic Computer

The Electronic Numerical Integrator and Computer (ENIAC) was developed in 1946 by John Mauchly and J. Presper Eckert. It was one of the first general-purpose electronic computers. ENIAC used thousands of vacuum tubes, occupied a large room, and performed calculations much faster than earlier machines.



UNIVAC

In 1951, UNIVAC (Universal Automatic Computer) became one of the first commercial computers. It could process both numbers and text, making it useful for business and government work. UNIVAC was an important step in bringing computers into practical use.



Tech Fact

Did You Know?

Univac had 5,200 vacuum Tubes

Features of First Generation Computers

- Uses Vacuum Tubes
- Very Expensive
- Huge in size
- Consumes lot of Energy

Second Generation Computers (1950s)

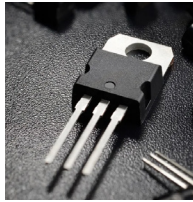
The invention of the transistor replaced bulky vacuum tubes and made computers smaller and more reliable. During this period, magnetic tapes were used for storing data. These computers worked faster, consumed less electricity, and required less maintenance than first-generation computers.



Features of Second Generation Computers

- Used transistors instead of vacuum tubes
- Smaller and more reliable
- Faster processing speeds
- Used magnetic tapes for data storage

Transistors
2nd Gen



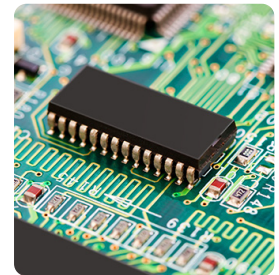
Vacuum Tube
1st Gen



Magnetic Tapes

Third Generation Computers (1960s)

The third generation introduced Integrated Circuits (ICs), which contained many electronic components on a single chip. This innovation greatly increased the speed and storage capacity of computers while reducing their size. Computers became more dependable and affordable, making them useful for businesses, schools, and scientific research.



Features of Third Generation Computers

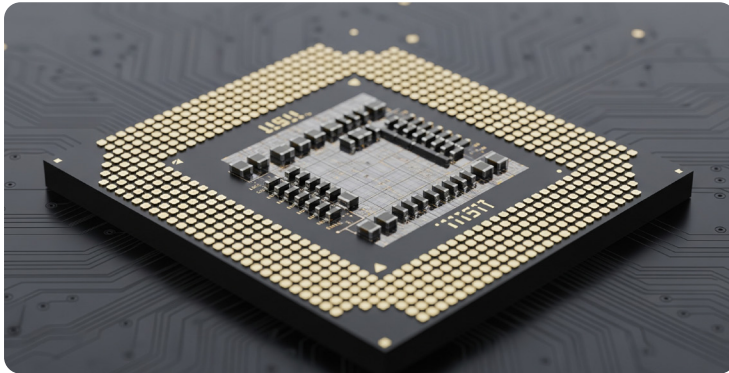
- Built using Integrated Circuits (ICs)
- Faster and more accurate
- Smaller in size
- More reliable and affordable

Fourth Generation Computers (1970s)

The invention of the microprocessor marked the beginning of the fourth generation. A microprocessor contains thousands of tiny electronic components on a single chip, making computers compact and highly efficient. Personal computers became popular during this generation, allowing people to use computers at home, in schools, and in offices. Modern operating systems with graphical interfaces also made computers easier to use.

Features of Fourth Generation Computers

- Based on microprocessor technology
- Small and portable
- High-speed processing
- User-friendly operating systems



Microprocessor



IBM PC



Tech Fact

In **1981**, IBM introduced the **IBM Personal Computer (IBM PC)**, making computers more suitable for homes, schools, and offices. It was powered by the **Intel 8088** microprocessor and became one of the most popular personal computers of its time. The IBM PC set a new standard for desktop computers, and many modern PCs are based on its design. Its success played an important role in the widespread use of personal computers around the world.



Fifth Generation Computers (Present)

The fifth generation represents the latest stage in the evolution of computers. These computers use Artificial Intelligence (AI) and advanced technologies that enable them to learn, recognise speech, understand languages, and solve complex problems. They are used in smart devices, robots, healthcare, education, space research, and many other fields. Scientists continue to improve these computers to make them even smarter and more capable.

Features of Fifth Generation Computers

- Uses Artificial Intelligence (AI)
- Extremely fast and efficient
- Supports voice and image recognition
- Used in smart devices and robotics



Modern Computers

Today's computers are available in different shapes and sizes to meet different needs.

Desktop Computers

A desktop computer is designed to stay on a table or desk. It usually consists of a monitor, CPU, keyboard, and mouse. Desktop computers are powerful and are commonly used in schools, offices, homes, and businesses. They are suitable for tasks such as studying, designing, programming, and gaming.



Laptop

A laptop is a portable computer that combines all the essential parts into one compact device. It runs on a rechargeable battery and can be carried easily from one place to another. Laptops are widely used by students, teachers, professionals, and travellers because they are convenient and easy to use.



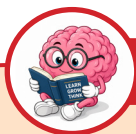
Tablet

A tablet is a lightweight computer with a touchscreen display. It is operated using fingers or a stylus instead of a keyboard and mouse. Tablets are commonly used for reading e-books, watching educational videos, attending online classes, drawing, and browsing the internet.



Smartphone

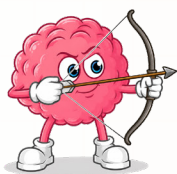
A smartphone is a small handheld computer that combines communication and computing in one device. Besides making calls and sending messages, it allows users to browse the internet, play games, take photographs, watch videos, attend online classes, and use thousands of useful applications. Smartphones have become an essential part of everyday life.



Quick Recap

- Early people used fingers, pebbles, and shells for counting.
- The Abacus was one of the earliest calculating devices.
- Blaise Pascal invented the Pascaline, the first mechanical calculator.
- Charles Babbage is known as the Father of the Computer.
- Ada Lovelace is called the first computer programmer.
- Herman Hollerith invented the Tabulating Machine.
- Computers have evolved through five generations.
- Modern computers are smaller, faster, and smarter than ever before.





TEST YOUR SKILLS



A. Choose the correct option

1. Which of these is known as the first calculating device?

- | | | | |
|-----------|-----|---------------|-----|
| a. Abacus | () | b. Smartphone | () |
| c. Tablet | () | d. Calculator | () |

2. Who is known as the Father of the Computer?

- | | | | |
|--------------------|-----|---------------------|-----|
| a. Blaise Pascal | () | b. Herman Hollerith | () |
| c. Charles Babbage | () | d. Lady Ada | () |

3. Which machine used punched cards to store data?

- | | | | |
|--------------|-----|-----------------------|-----|
| a. Pascaline | () | b. Tabulating Machine | () |
| c. Abacus | () | d. Laptops | () |

4. Which generation introduced the microprocessor?

- | | | | |
|----------|-----|-----------|-----|
| a. First | () | b. Second | () |
| c. Third | () | d. Fourth | () |

5. Artificial Intelligence is mainly used in the which generation?

- | | | | |
|----------|-----|-----------|-----|
| a. Third | () | b. Fourth | () |
| c. Fifth | () | d. Second | () |

B. Fill in the Blanks

Word Bank: Pascaline, AI, Vacuum Tubes, Abacus, Ada Lovelace

- _____ is one of the oldest calculating devices.
- The first mechanical calculator was called _____.
- First-generation computers used _____.
- _____ is known as the first computer programmer.
- Fifth-generation computers use _____.



C. State True or False

1. The Abacus was invented after the computer. _____
2. Charles Babbage designed the Analytical Engine.. _____
3. Transistors replaced vacuum tubes in the second generation. _____
4. Microprocessors were introduced in the third generation. _____
5. Smartphones are examples of modern computers. _____

D. Answer The Following Questions

1. Why did early people use pebbles and shells for counting?
2. Why is Charles Babbage called the Father of the Computer?
3. Write any two features of second-generation computers.
4. What is a microprocessor?
5. Name any two modern computers and write one use of each.

E. Competency-Based Questions

- a. Rahul wants to solve simple addition problems without using a calculator. Which early calculating device would you suggest? Why?
- b. Your school is replacing old desktop computers with new AI-enabled computers. Write two advantages of the new computers over the old ones.

F. Let's Explore

Talk to an elderly family member and ask how computers have changed during their lifetime. Write any three changes they noticed.

1. _____
2. _____
3. _____

G. Let's Get Better

Fill the table below using the information you learned in this chapter:

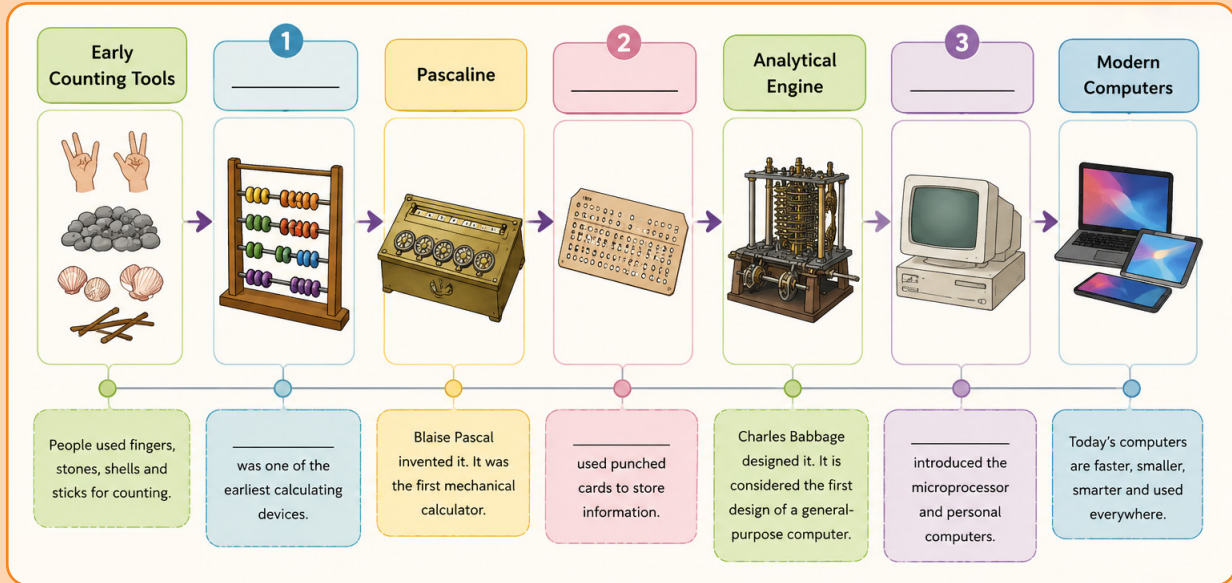
Inventor	Invention	Year





Tech Practice

Fill in the missing inventions in the correct order:



Lets Solve

Word Search

WORDS TO FIND

- ABACUS
- PASCALINE
- ENIAC
- UNIVAC
- MARKONE
- BABBAGE
- LOVELACE
- HOLLERITH
- VLSI
- COMPUTER

P	A	Q	V	P	I	S	E	U	D	T	B
X	W	V	H	L	U	F	E	N	G	P	P
F	V	B	S	C	S	C	M	I	I	Z	P
X	V	B	A	L	E	I	B	V	R	A	O
J	X	B	E	N	O	K	R	A	M	P	C
B	A	B	B	A	G	E	S	C	V	U	I
P	A	S	C	A	L	I	N	E	T	Z	Q



Teacher's Note

- *Help students relate early counting tools with the modern devices they use every day.*
- *Conduct a simple classroom activity to compare the size, speed, and uses of computers from different generations.*



Advanced Features of Word 2016



Let's Warm Up the Brain

Learning Goals

- Spelling & Grammar
- Thesaurus
- Find & Replace Text
- Page Formatting
- Column Formatting
- Paragraph Formatting
- Mail Merge

Ayush is preparing a school notice on the computer. He wants everyone to understand his message clearly. However, he made a few spelling mistakes while typing.

Why do you think it is important to correct spelling mistakes before sharing a document?

In this chapter, you will learn about some important advanced features of Microsoft Word, such as checking spelling and grammar, using the Thesaurus, finding and replacing text, formatting pages, arranging text into columns, and improving paragraph appearance.

Spelling and Grammar

Have you ever noticed that Microsoft Word sometimes underlines certain words or sentences while you are typing? These coloured lines are helpful hints that tell you something in your document may need correction. Instead of searching for every mistake yourself, Word automatically checks your writing and alerts you whenever it finds a possible spelling or grammar error. The Spelling and Grammar feature helps you create documents that are accurate and professional. It checks the words you type and suggests corrections whenever it detects mistakes.

Generally, Word uses different coloured wavy lines to indicate these errors:

- A red wavy line appears below a word that may have been spelled incorrectly.
- A blue wavy line appears below a sentence or phrase that may contain a grammar mistake or incorrect word usage.

You can review these suggestions and either accept the correction or ignore it if the word is correct.



Checking Spelling and Grammar

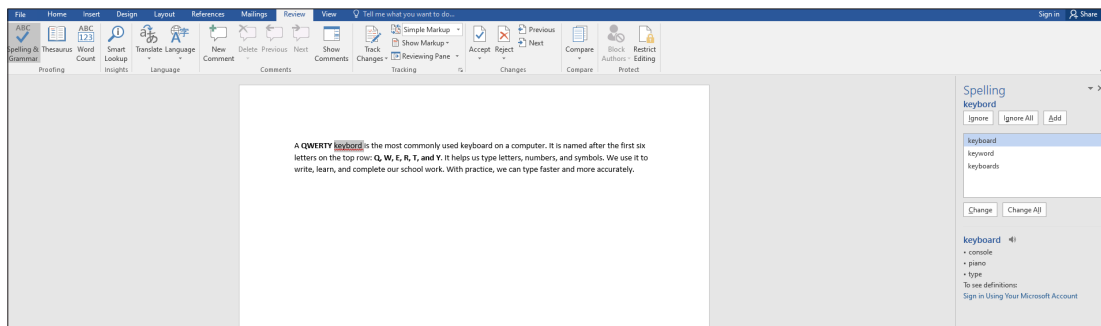
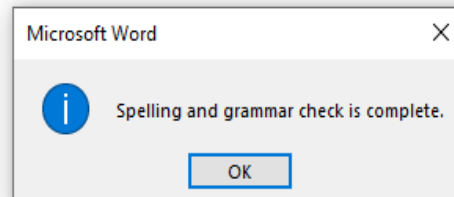
Step 1: Open the document you want to check.

Step 2: Click the Review tab on the Ribbon.

Step 3: In the Proofing group, click Spelling & Grammar.

Step 4: Word starts checking the document from the current cursor position. If it finds a mistake, a Spelling or Editor pane appears with suggested corrections.

Step 5: Review the suggestions and choose the appropriate option.



You may use the following options:

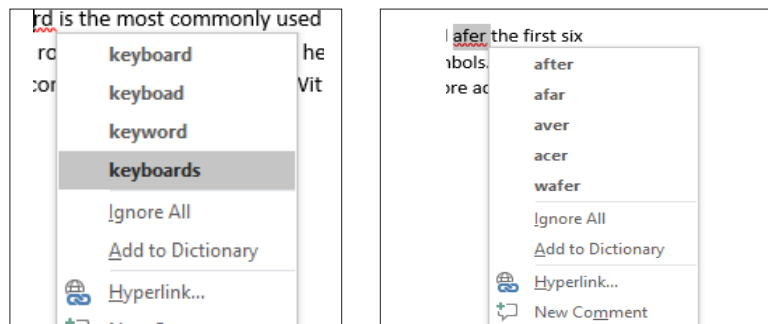
Change: Replaces the selected word with the suggested spelling.

Change All: Replaces every occurrence of the same incorrect word throughout the document.

Ignore Once: Skips the current suggestion without making any changes.

Ignore All: Ignores all occurrences of the same word in the document.

Add to Dictionary: Adds a correctly spelled word, such as a person's name or place, to Word's dictionary so it is not marked as incorrect again.



Thesaurus

The Thesaurus is a built-in collection of synonyms and related words. It allows you to improve your writing by choosing words that better match the meaning of your sentence. Using suitable synonyms also makes your document richer and more attractive to read.

Using the Thesaurus

Step 1: Select the word for which you want to find a synonym.

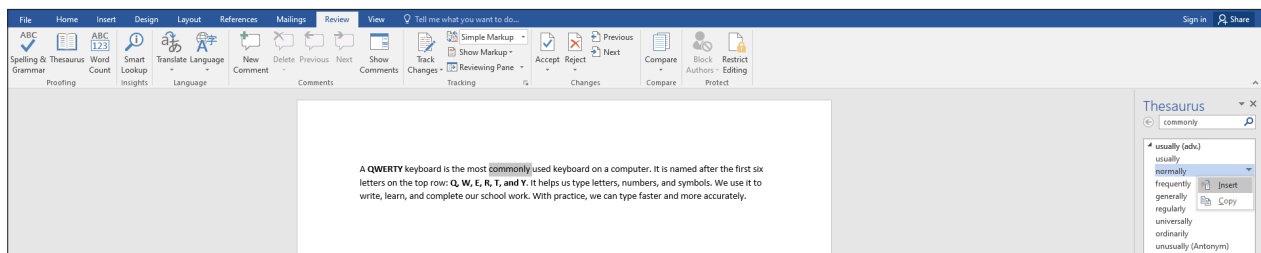
Step 2: Click the Review tab.

Step 3: In the Proofing group, click Thesaurus.

Step 4: The Thesaurus pane opens on the side of the window and displays a list of similar words.

Step 5: Click the drop-down arrow beside the desired synonym and choose Insert or Copy, depending on how you want to use it.

The selected synonym replaces or can be inserted into your document, making your writing more effective.



A QWERTY keyboard is the most normally used keyboard on a computer. It is named after the first six letters on the top row: Q, W, E, R, T, and Y. It helps us type letters, numbers, and symbols. We use it to write, learn, and complete our school work. With practice, we can type faster and more accurately.



Tech Fact

Press Shift + F7 to open the Thesaurus quickly.
A synonym is a word that has the same or nearly same meaning as other word.



Find and Replace Text

The Find feature helps you search for a specific word, phrase, or number in a document. The Replace feature not only searches for the required text but also replaces it with new text automatically. These tools are very useful while editing reports, assignments, letters, and project files.

Finding The Text

Step 1: Open the document.

Step 2: Click the Home tab.

Step 3: In the Editing group, click Find.

Step 4: The Navigation pane appears on the left side of the window.

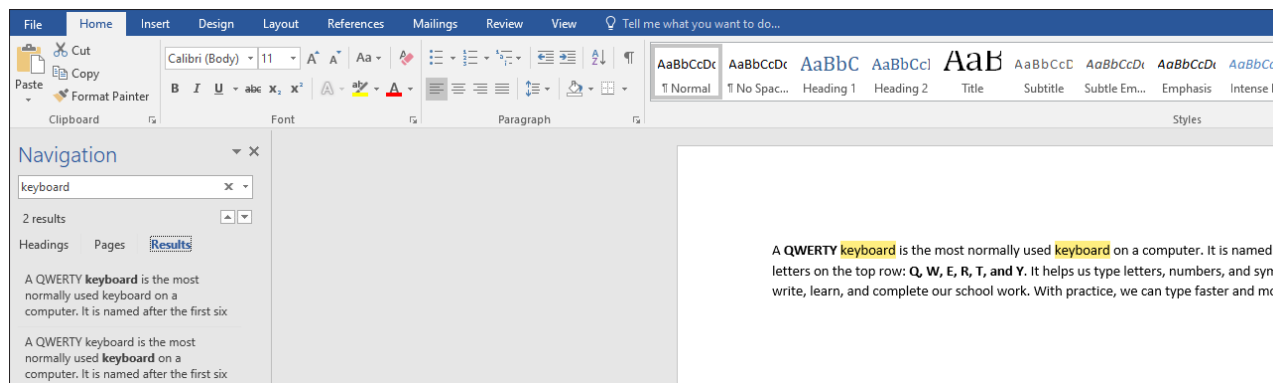
Step 5: Type the word or phrase you want to search for in the search box.

Step 6: Word highlights every occurrence of the searched word in the document.

A list of matching results is also displayed in the Navigation pane.

Step 7: Click any result in the Navigation pane to jump directly to that location in the document.

Using the Find command saves time, especially when working with lengthy documents containing many pages.



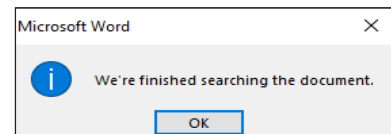
Replacing The Text

Step 1: Click the Home tab.

Step 2: In the Editing group, click Replace.

Step 3: The Find and Replace dialog box opens.

Step 4: Type the word or phrase you want to search for in the Find what box.



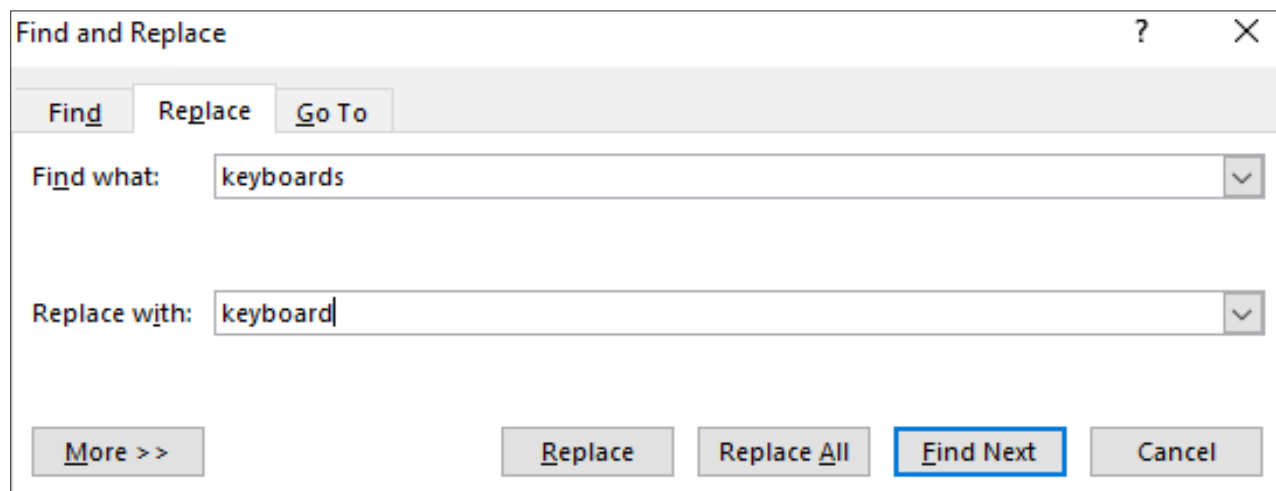
Step 5: Type the new word or phrase in the Replace with box.

Step 6: Click Find Next to locate the first occurrence of the word.

Step 7: Choose one of the following options:

Replace: Replaces only the currently selected occurrence and moves to the next one.

Replace All: Replaces every occurrence of the searched word throughout the document at once.



Tech Fact

Ctrl + F : Open the Find tool.

Ctrl + H : Open the Find and Replace dialog box.

Step 8: After the replacement is complete, Word displays a message showing how many replacements were made.



Step 9: Click OK, and then click Close to exit the dialog box.

Page Formatting

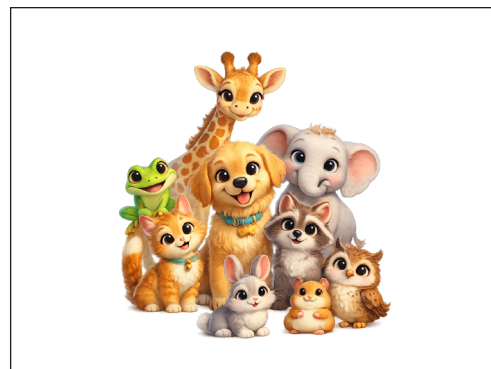
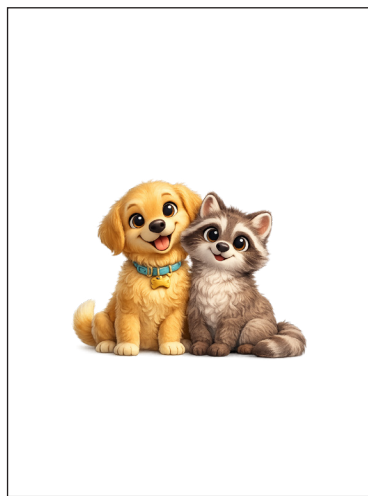
When you create a document, it is important to arrange the page neatly so that it looks attractive and is easy to read. The process of setting up the page by adjusting its size, direction, margins, and layout is called page formatting.

Changing Page Orientation

Page orientation refers to the direction in which a page is displayed or printed. Microsoft Word offers two types of page orientations.

Portrait: The page is taller than it is wide. It is the default orientation in Microsoft Word and is suitable for letters, essays, reports, and assignments.

Landscape: The page is wider than it is tall. It is useful for certificates, tables, charts, drawings, and wide pictures.



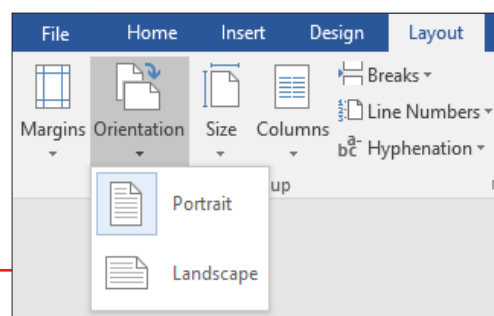
To change the page orientation:

Step 1: Open your document in Microsoft Word.

Step 2: Click the Layout tab.

Step 3: In the Page Setup group, click Orientation.

Step 4: Select Portrait or Landscape from the drop-down list.



Setting Paper Size

The paper size determines the height and width of a page. Selecting the correct size ensures that your document prints correctly without cutting off text or images.

Some commonly used paper sizes are:

A4: Commonly used for school projects, office work, and reports.

Letter: Used in many countries for official and personal documents.

Legal: Used for legal papers and long documents.

Among these, A4 is the standard paper size used in most schools.

To change the paper size:

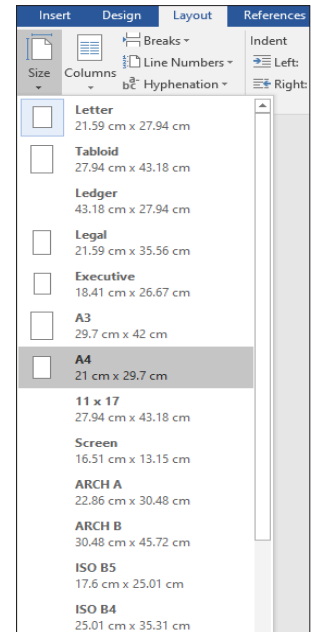
Step 1: Click the Layout tab.

Step 2: In the Page Setup group, click Size.

Step 3: A list of available paper sizes appears.

Step 4: Select the paper size you want to use.

The document will now be adjusted according to the selected paper size.



Inserting a Page Break

A Page Break tells Microsoft Word to end the current page and automatically start the next page from the position of the cursor. It helps keep your document organised.

To insert a page break:

Step 1: Place the cursor where you want the next page to begin.

Step 2: Click the Layout tab.

Step 3: In the Page Setup group, click Breaks.

Step 4: Under Page Breaks, select Page.

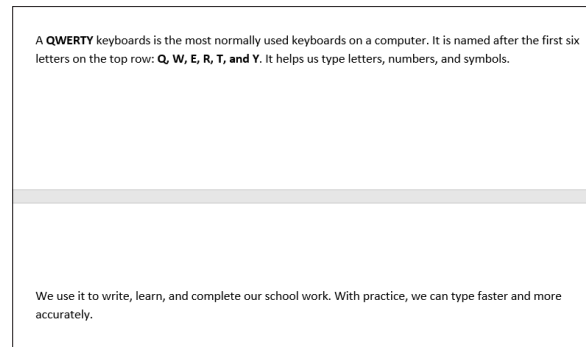
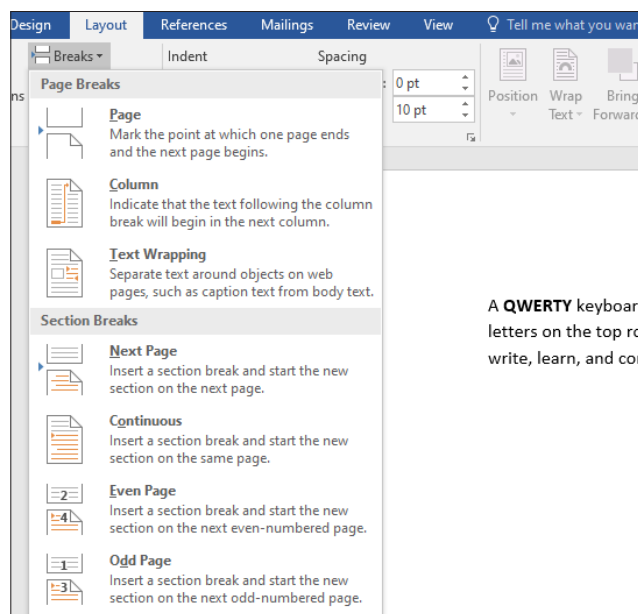
A new page is inserted, and the cursor automatically moves to the beginning of the next page.



Tech Fact

You can quickly insert a page break by pressing Ctrl + Enter on the keyboard.





Column Formatting

Column formatting is commonly used in newspapers, magazines, brochures, newsletters, and pamphlets. It helps organise information neatly and allows readers to read long passages of text more comfortably. Word automatically adjusts the text from one column to the next, giving the document a professional look.

You can choose from different column layouts, such as one, two, three, or more columns, depending on the type of document you are creating.

To create columns in a document:

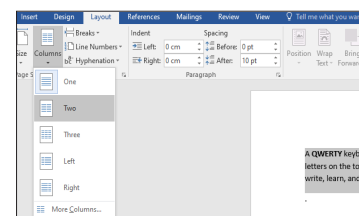
Step 1: Select the text that you want to arrange into columns. To apply columns to the entire document, leave the text unselected.

Step 2: Click the Layout tab on the Ribbon.

Step 3: In the Page Setup group, click Columns.

Step 4: A list of column layouts appears.

Step 5: Choose the required option, such as One, Two, Three, Left, or Right.



A **QWERTY** keyboard is the most commonly used keyboard on a computer. It is named after the first six letters on the top row: **Q, W, E, R, T, and Y**. It helps us type letters, numbers, and

symbols. We use it to write, learn, and complete our school work. With practice, we can type faster and more accurately.

Paragraph Formatting

In Microsoft Word, paragraph formatting helps you organise text neatly by adjusting its alignment, spacing, indentation, and overall appearance. Proper paragraph formatting makes a document more attractive and easier to understand. It improves readability by creating enough space between lines and paragraphs, making the document look clean and well organised.

Line Spacing

Line spacing is the amount of vertical space between one line of text and the next within the same paragraph. Increasing line spacing makes the document appear more open and comfortable to read, while decreasing it allows more text to fit on a page. Depending on the purpose of the document, Word offers several line spacing options such as 1.0 (Single), 1.15, 1.5, and 2.0 (Double).

To change line spacing:

Step 1: Select the paragraph or text whose line spacing you want to change.

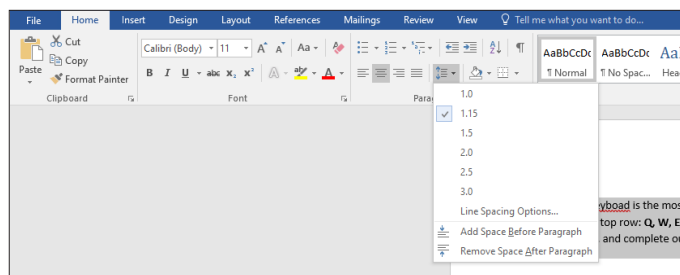
Step 2: Click the Home tab.

Step 3: In the Paragraph group, click the Line and Paragraph Spacing button.

Step 4: A list of spacing options appears.

Step 5: Click the required spacing option, such as 1.0, 1.15, 1.5, or 2.0.

The selected line spacing is applied immediately.



A QWERTY keyboad is the most commonly used keyboard on a computer. It is named after the first six letters on the top row: **Q, W, E, R, T, and Y**. It helps us type letters, numbers, and symbols. We use it to write, learn, and complete our school work. With practice, we can type faster and more accurately.

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Paragraph Spacing

Paragraph spacing separates one paragraph from another without pressing the Enter key multiple times. It keeps the document neat and ensures consistent spacing throughout the document. Using paragraph spacing is considered a better formatting practice than adding blank lines manually.

To change paragraph spacing:

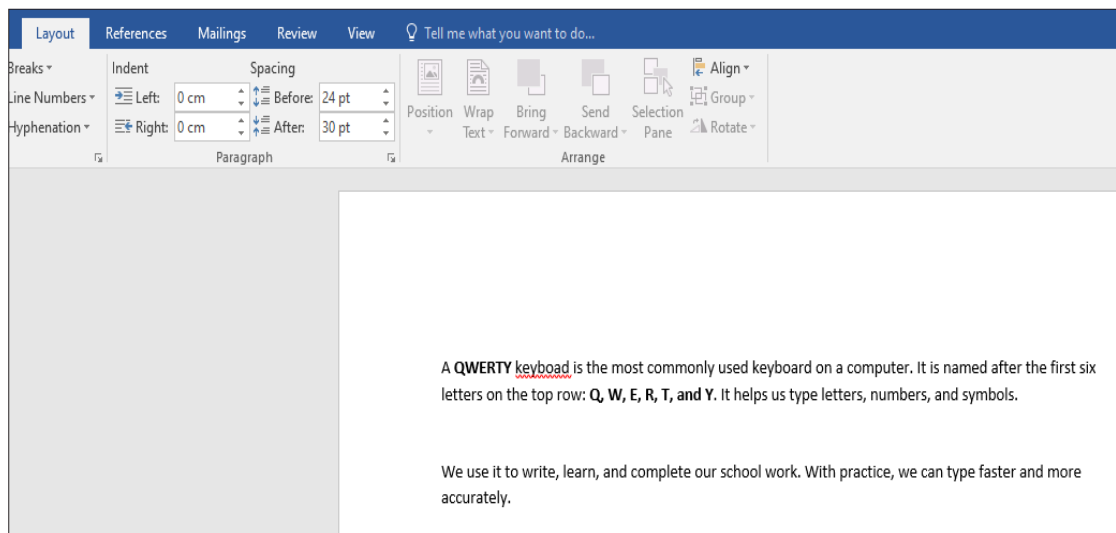
Step 1: Select the paragraph that you want to modify.

Step 2: Click the Layout tab.

Step 3: In the Paragraph group, locate the Spacing section.

Step 4: Increase or decrease the values in the Before and After boxes according to your requirement.

The spacing between paragraphs is updated instantly.



Tech Fact

Instead of pressing the Enter key several times to create gaps, use Paragraph Spacing to maintain a neat and professional document.

Mail Merge

In Mail Merge, the main content of the document remains the same, while details such as the recipient's name, address, or phone number change automatically for each copy. This makes mail merge a fast and accurate way to prepare documents for many people.



Mail Merge mainly works with three parts:

Main Document: It contains the common text that will remain the same for every recipient. Spaces are left for merge fields where personal information will appear.

Data Source: It stores the details of all the recipients, such as their names, addresses, phone numbers, or email addresses.

Merged Document: It is the final document created after combining the main document with the data source. Each copy contains the same message but different recipient information.

Creating the Main Document

Suppose your school is organising an Annual Science Exhibition, and you want to invite parents, teachers, and guests. Since the invitation message is the same for everyone, you only need to create the main document once.

To create the main document:

Step 1: Open Microsoft Word and type the invitation letter.

To,

Dear Riya,

I hope you are doing well. I am happy to invite you to my birthday party on **10 August 2026 at 5:00 p.m.** at my home. We will have games, music, delicious food, and lots of fun. Please come with your parents and make my birthday special with your presence.

I will be waiting for you.

Thanking you,

Ananya

Step 2: Click the Mailings tab.

Step 3: In the Start Mail Merge group, click Start Mail Merge.

Step 4: A drop-down list appears. Select Letters.

Your document is now ready to be connected with a list of recipients.

Creating a Data Source

The next step is to create a list containing the details of all the people who will receive the invitation. This list is known as the Data Source. The data source can include information such as: First Name, Last Name, Address, City, State, Pin Code, Phone Number, Email Address.

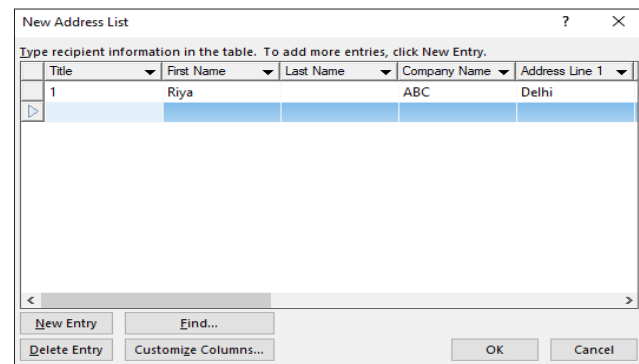
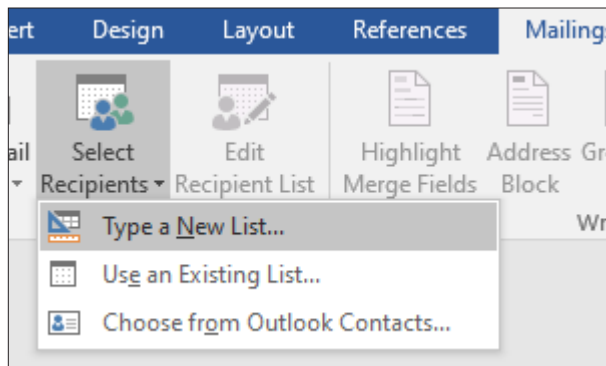
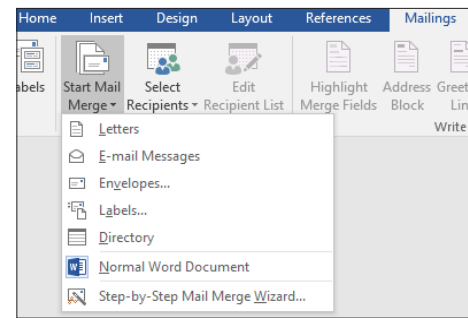


To create a data source follow these steps:

Step 1: Click on the Mailings tab.

Step 2: In the Start Mail Merge group, click on Select Recipients.

Step 3: From the drop-down menu, select Type a New List. The New Address List dialog box appears on the screen.



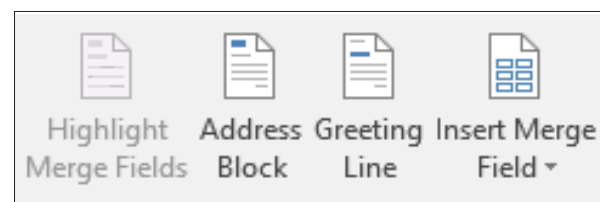
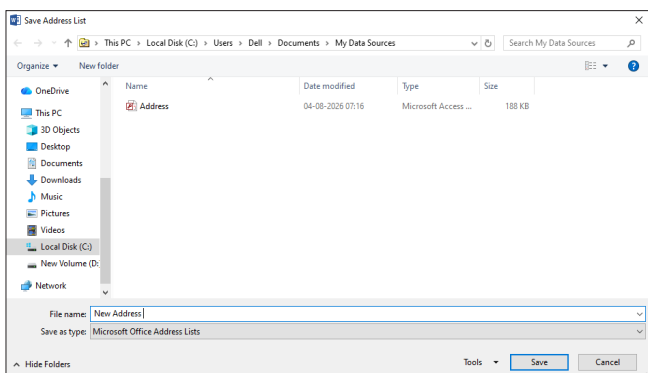
Step 4: Enter the details of the first recipient in the appropriate fields, such as First Name, Last Name, Address, City, State, PIN Code, Phone Number, and E-mail Address.

Step 5: To add another recipient, click the New Entry button and enter the required details. If you want to remove a recipient, click the Delete Entry button.

Step 6: After entering all the recipient details, click the OK button. A Save Address List dialog box appears. In that box, type a suitable name for your address list.

Step 7: Click the Save button.

You will notice that merge fields such as <<AddressBlock>> or other fields can now be inserted into your document.



Merging the Main Document and Data Source

After creating both the main document and the data source, the final step is to combine them. During this process, Word automatically inserts each recipient's information into the appropriate merge fields and creates a separate copy for every person.

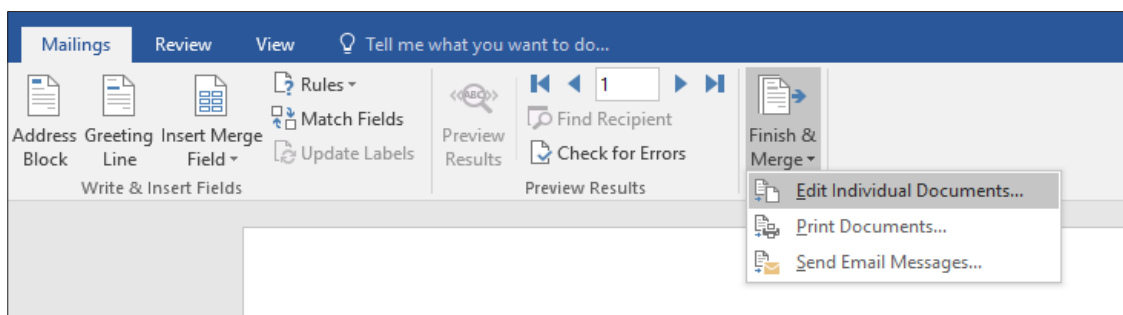
To merge the documents:

Step 1: Click the Mailings tab.

Step 2: In the Finish group, click Finish & Merge.

Step 3: Select Edit Individual Documents from the menu.

A Merge to New Document dialog box appears.



Step 4: Choose one of the following options:

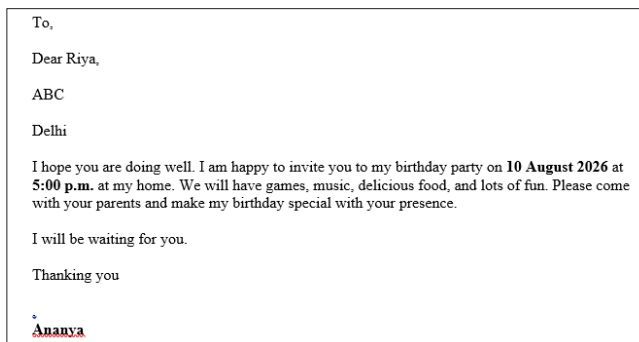
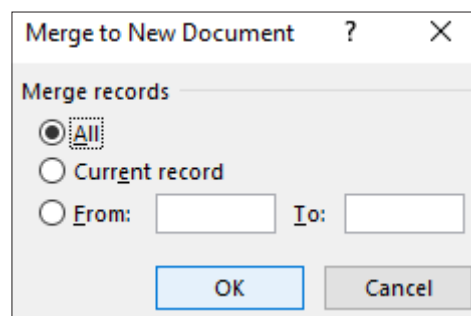
All: Creates documents for all recipients.

Current Record: Creates a document only for the selected recipient.

From: To: Creates documents for a specific range of recipients.

Step 5: Click OK.

Microsoft Word creates a new document containing personalised letters for all the selected recipients.





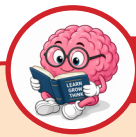
Activity

Design an Invitation:

Create an invitation in Microsoft Word for your school's Annual Function. Use the following features:

- Spell Check
- Thesaurus (change at least two words using synonyms)
- Page Orientation
- Paragraph Spacing

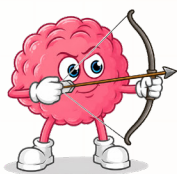
Save your document with the name Annual Function Invitation.



Quick Recap

- *Spell Check helps correct spelling and grammar mistakes.*
- *Thesaurus provides synonyms for selected words.*
- *Find searches for a word or phrase in a document.*
- *Replace changes one word or phrase with another.*
- *Page Formatting improves the appearance of a document.*
- *Portrait and Landscape are the two page orientations.*
- *A Page Break starts a new page in a document.*
- *Column Formatting arranges text into columns.*
- *Paragraph Formatting changes the appearance of paragraphs.*
- *Line Spacing controls the space between lines.*
- *Paragraph Spacing controls the space between paragraphs.*
- *Mail Merge creates personalised documents for multiple recipients.*





TEST YOUR SKILLS



A. Choose the correct option

1. Which feature helps create personalised letters for many people?
a. Find ☐ b. Mail Merge ☐
c. Thesaurus ☐ d. Replace ☐
2. Which tab contains the Replace command?
a. Home ☐ b. Insert ☐
c. View ☐ d. Layout ☐
3. Which feature helps you find words with similar meanings?
a. Spell Check ☐ b. Replace ☐
c. Mail Merge ☐ d. Thesaurus ☐
4. Which page orientation is suitable for wide tables and charts?
a. Portrait ☐ b. Landscape ☐
c. Square ☐ d. Circle ☐
5. Which shortcut key opens the Spelling and Grammar tool?
a. Ctrl + F ☐ b. Shift + F7 ☐
c. F7 ☐ d. Alt + F7 ☐

B. Fill in the Blanks

Word Bank: Mail Merge, Landscape, Thesaurus, Replace, Paragraph Spacing

1. The _____ tool provides synonyms for selected words.
2. _____ changes one word or phrase into another.
3. _____ orientation is suitable for certificates and wide documents.
4. _____ controls the space between paragraphs.
5. _____ is used to create personalised letters for many recipients.



C. State True or False

1. The Find command changes the selected word automatically. _____
2. Line spacing controls the distance between lines of text. _____
3. A Page Break starts a new page in a document. _____
4. Thesaurus checks grammar mistakes. _____
5. Mail Merge combines a main document with a data source. _____

D. Answer The Following Questions

1. What is the purpose of the Thesaurus tool?
2. Explain the steps to use the Find command.
3. What is Page Orientation?
4. How do you insert a Page Break?
5. What is Mail Merge? Mention any two of its uses.

E. Competency-Based Questions

- a. Priya has completed her English project in Microsoft Word. She notices that some words are repeated many times. Which feature can help her improve her writing? Explain how she can use it.
- b. Your school wants to send the same invitation to all parents, but each invitation should contain a different parent's name and address. Which Microsoft Word feature should be used? Explain the basic steps involved.

F. Let's Explore

Create a one-page notice in Microsoft Word for your school's Science Exhibition.

After typing the notice:

- Check the spelling and grammar.
- Replace the word students with participants.
- Use the Thesaurus to replace one repeated word with a suitable synonym.
- Change the page to Landscape orientation.

Save the document with the name Science Exhibition Notice.

G. Let's Get Better

Your teacher wants to send appreciation certificates to all the winners of a competition. How can Mail Merge help prepare these certificates quickly?





Tech Practice

Practice the following questions in MS Word:

1. Type the paragraph below in Microsoft Word and correct any spelling or grammar mistakes using the Spelling and Grammar tool.
2. Use the Find command to search for the word computer and replace it with laptop wherever it appears.
3. Select any two words and use the Thesaurus to replace them with suitable synonyms.
4. Change the document to Landscape orientation and apply 1.5 line spacing.
5. Create a small Mail Merge recipient list with five names and prepare invitation letters for all recipients.

H. Let's Solve

Write the keyboard shortcuts for the following:

Spelling and Grammar

Find Text

Replace Text

Open Thesaurus

Insert a Page Break

Teacher's Note

- Demonstrate the use of Spelling & Grammar, Find, Replace, and Mail Merge using simple classroom examples.
- Encourage students to format a document by changing page orientation, spacing, and columns to improve its appearance.



Formatting a Presentation



Let's Warm Up the Brain

Learning Goals

- Applying Themes
- Working with Slide Master
- Tweaking Background
- Smart Art
- Slide Animation & Transitions

Imagine you are creating a presentation about "Our Solar System." Which slide layout would you choose for each task?

Task	Your Choice
Write title & subtitle	
Show a large picture with heading	
Compare Earth and Mars	
Add a title with a short list of facts	
Display a picture with description	

Applying Themes

A theme is a ready-made design that gives your presentation a professional and attractive appearance. It includes matching colours, fonts, background styles and effects that make every slide look neat and uniform. Instead of designing each slide separately, you can apply a theme to the entire presentation with just a few clicks. Using a theme saves time and helps create presentations that are easy to read and visually appealing.

Steps to Apply a Theme

Step 1: Open your presentation in Microsoft PowerPoint.

Step 2: Click the Design tab on the Ribbon. The Design tab contains tools that help you improve the overall appearance of your presentation, such as Themes, Variants, Designer and Background options.

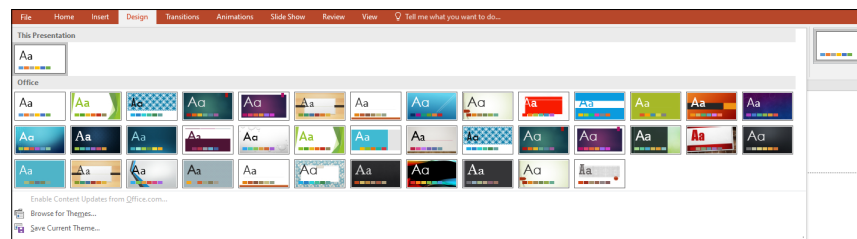
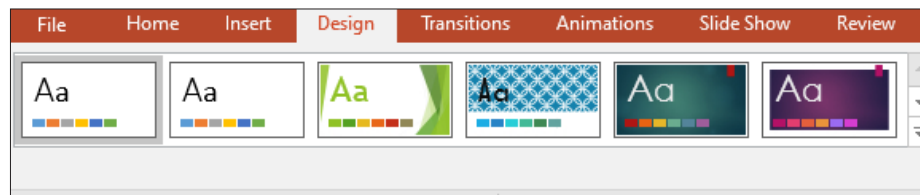
Step 3: In the Themes group, click the More drop-down arrow. A gallery displaying all the available built-in themes opens on the screen. Scroll through the gallery to view the different theme designs provided by PowerPoint.

Step 4: Move the mouse pointer over any theme in the gallery without clicking it. PowerPoint displays a Live Preview, allowing you to see how the selected theme will look on all the slides in your presentation.



Step 5: After selecting a suitable theme, click on it. The chosen theme is immediately applied to every slide in the presentation. All the slides now use the same colours, fonts, background style and visual effects.

Step 6: Save your presentation to keep the changes made by the newly applied theme.



After applying a theme and selecting a suitable design idea, your presentation becomes more attractive and well-organised. The slides have a consistent appearance, making the information easier to read and understand. You can change the theme or choose another design suggestion at any time without affecting the content of your presentation.

Modifying Themes

After applying a theme, you may want to give your presentation a different look without changing the entire design. Microsoft PowerPoint allows you to modify various elements of a theme, such as its colours, fonts, effects and background styles. These changes help you create a presentation that matches the topic while maintaining a consistent appearance on every slide.



Working with Slide Master

The Slide Master controls the overall design and layout of a presentation. It stores information such as the theme, fonts, colours, background, placeholders and the position of different objects. Any change made to the Slide Master is automatically reflected on all the slides linked to that master. This helps maintain a consistent appearance throughout the presentation and saves both time and effort.

Steps to open Slide Master

Step 1: Open the presentation in Microsoft PowerPoint.

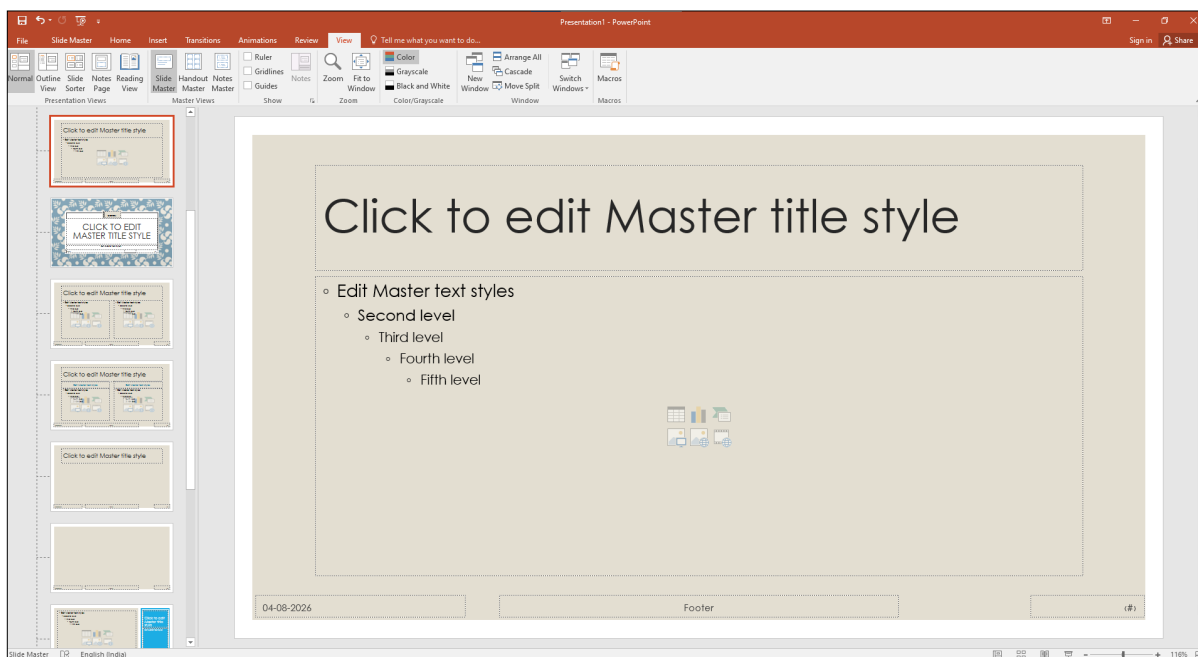
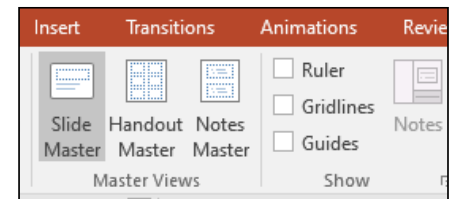
Step 2: Click the View tab on the Ribbon. The View tab contains different options for viewing and editing presentations.

Step 3: In the Master Views group, click Slide Master. The presentation switches to Slide Master View.

Step 4: In the left pane, a list of slide masters and their related layouts appears. Click the topmost Slide Master if you want the changes to be applied to all the slides in the presentation.

Step 5: Make the required changes, such as changing the font style, adding a picture or logo, modifying the background or repositioning placeholders.

Step 6: After completing the changes, click the Slide Master tab and select Close Master View. PowerPoint returns to the Normal View, and the changes are automatically applied to the slides.



Any modification made in the Slide Master is reflected on all the slides associated with it. This ensures that every slide has a uniform appearance and eliminates the need to edit each slide separately. Using Slide Master is especially useful for creating professional presentations with a consistent design.



Changing the Background

You can apply a background to a single slide or to all the slides in the presentation.

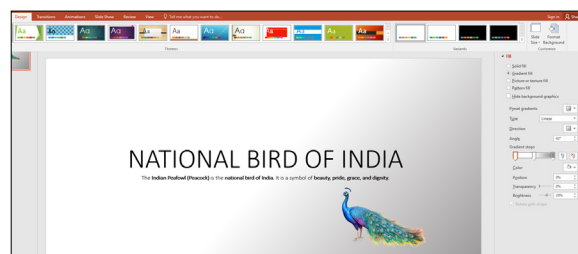
Steps to Change the Background

Step 1: Select the slide whose background you want to change. Click the Design tab on the Ribbon and, in the Customize group, click the Format Background option. The Format Background pane opens on the right side of the PowerPoint window.

Step 2: In the Format Background pane, choose the desired background option, such as Solid Fill, and then select a colour of your choice. The selected background is immediately displayed on the slide.

Step 3: If you want the same background on every slide, click the Apply to All button at the bottom of the Format Background pane. Otherwise, close the pane to keep the changes only on the selected slide.

After changing the background, the slides look more attractive and better suited to the presentation topic. Choosing a simple background with clear text helps the audience read and understand the information easily.



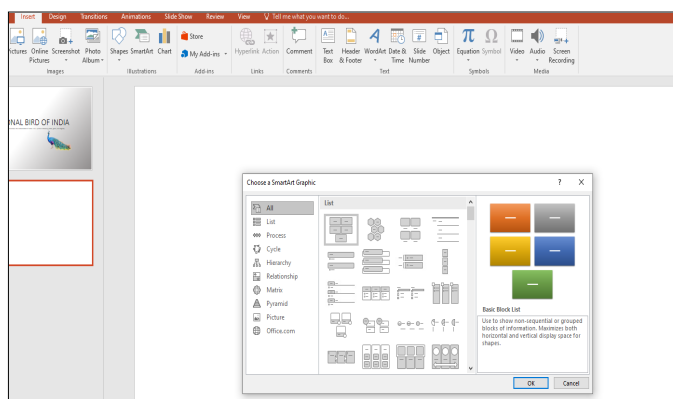
Inserting Smart Art

Microsoft PowerPoint provides SmartArt to present information in a visual and organised way. SmartArt uses different diagrams to display ideas, relationships, processes, cycles and hierarchies, making a presentation more attractive and easier to understand.

PowerPoint offers a variety of SmartArt layouts for different purposes. For example, you can use a Process layout to show steps in order, a Cycle layout to represent a repeating process, or a Hierarchy layout to display an organisational structure. Using the appropriate SmartArt graphic helps the audience understand information quickly and makes the presentation more interesting.

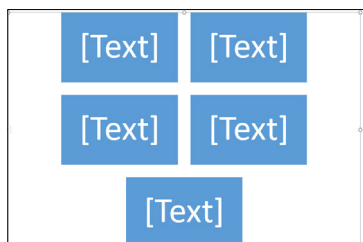
Steps to insert Smart Art in your presentation are as follows:

Step 1: Select the slide in which you want to insert a SmartArt graphic. Click the Insert tab on the Ribbon and then click the SmartArt button in the Illustrations group. The Choose a SmartArt Graphic dialog box appears on the screen.



Step 2: In the dialog box, select a category such as List, Process, Cycle, Hierarchy, Relationship, Matrix, Pyramid or Picture from the left pane. Choose the desired SmartArt layout from the middle pane. A preview and description of the selected layout are displayed on the right side of the dialog box.

Step 3: Click the OK button. The selected SmartArt graphic is inserted into the slide. Type the required text in the Text Pane or directly inside the SmartArt shapes to complete the graphic.



Animations

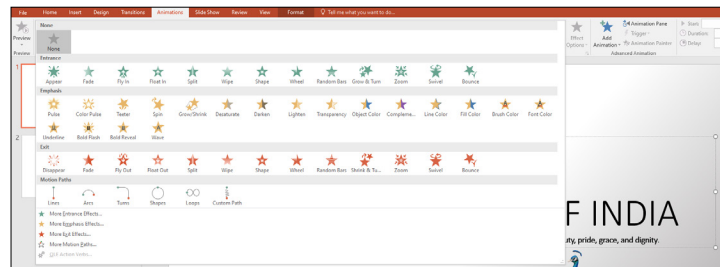
Animations are visual effects that can be applied to text, pictures, shapes and other objects on a slide. These effects make the objects appear, move or disappear while the presentation is being shown. Using animations makes a presentation more attractive and helps keep the audience interested.

Steps to apply animations are as follows:

Step 1: Select the text, picture or object on the slide to which you want to apply an animation effect.

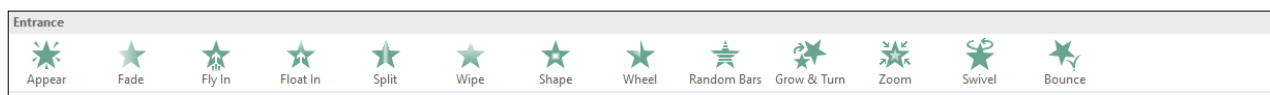
Step 2: Click the Animations tab on the Ribbon. The Animation group displays a collection of commonly used animation effects.

Step 3: Click the More button in the Animation group. A drop-down gallery containing various animation effects appears on the screen.

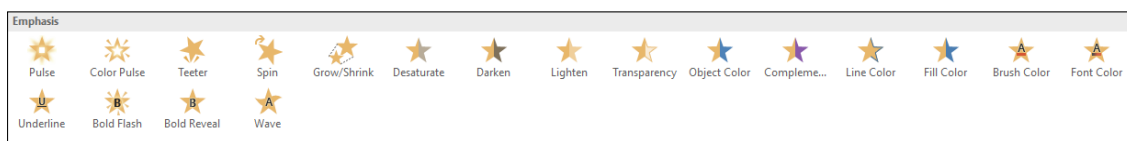


PowerPoint groups animation effects into four different categories. These categories are:

Entrance: These effects control how an object enters the slide. They are used to make text or objects appear in different styles when the presentation begins.

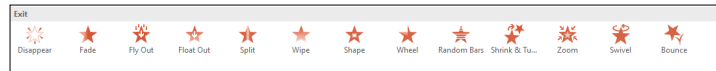


Emphasis: These effects are used to highlight important text or objects that are already present on the slide. They help attract the audience's attention without making the object disappear.

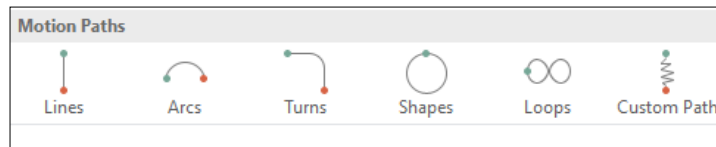


Exit: These effects control how an object leaves or disappears from the slide during the presentation.

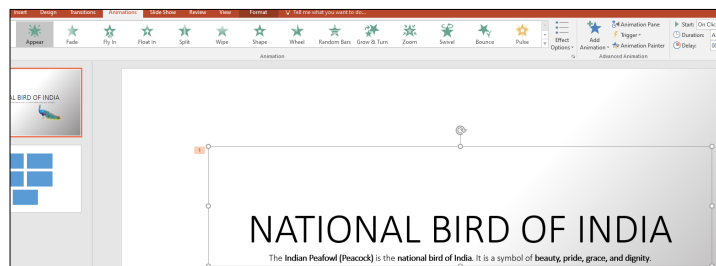




Motion Paths: These effects move an object from one place to another along a selected path, such as a line, curve, circle or other predefined pattern.



Step 4: Click the animation effect that you want to apply. A small numbered box appears beside the selected object, indicating that the animation has been successfully added and showing its order in the animation sequence.



Tech Fact

Instead of pressing the Enter key several times to create gaps, use Paragraph Spacing to maintain a neat and professional document.

Previewing an Animation

To check how an animation will appear during the slide show, you can preview it before presenting.

Step 1: Click the Animations tab on the Ribbon.

Step 2: In the Preview group, click the Preview button. PowerPoint plays the selected animation on the current slide so that you can see how it will look during the presentation.

Timing

PowerPoint provides different start options for animation effects. You can make an animation begin when you click the mouse, play together with another animation



or start automatically after the previous animation ends. You can also change the duration of an animation and rearrange its sequence whenever required.

Steps to Modify Animation Timing

Step 1: Click the Animations tab on the Ribbon.

Step 2: In the Timing group, click the Start drop-down arrow and choose any one of the following options:

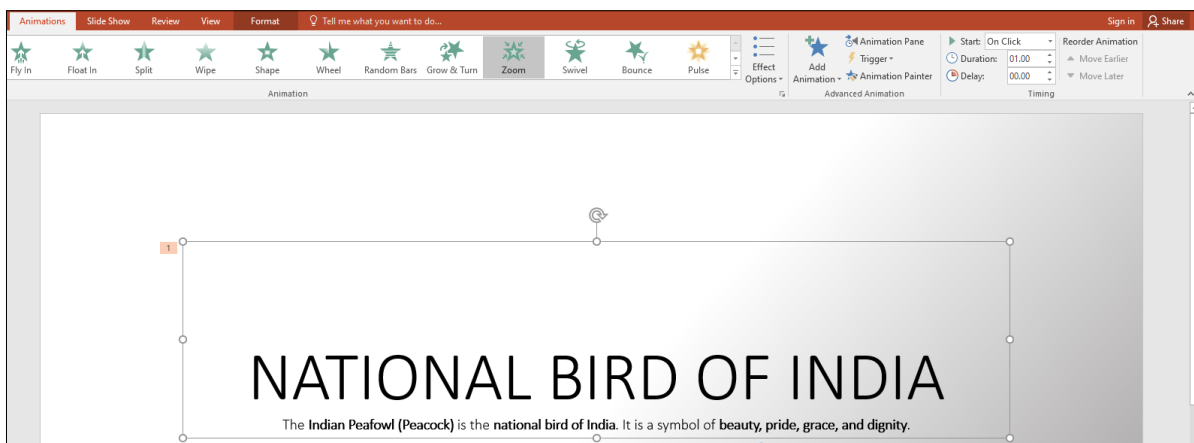
On Click: The animation starts when you click the mouse during the slide show.

With Previous: The animation starts at the same time as the previous animation effect.

After Previous: The animation starts automatically after the previous animation has finished.

Step 3: Enter the desired time in the Duration box to decide how long the animation effect should take to complete.

Step 4: Click Move Earlier or Move Later to change the order in which the animation effects are played.



Animation Pane

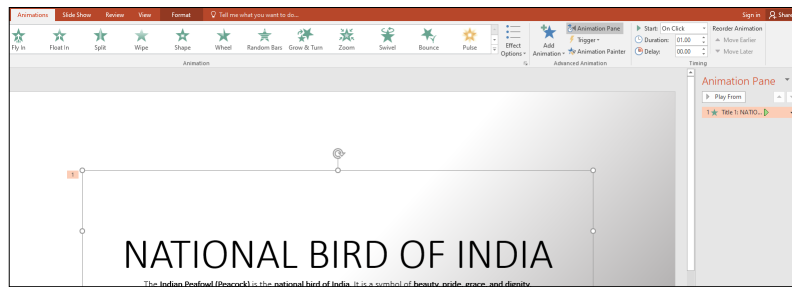
The Animation Pane displays all the animation effects that have been applied to the current slide. It helps you view the sequence of animations and makes it easier to organise them. You can also use this pane to change the timing, start options and duration of individual animation effects.

Steps to Open the Animation Pane

Step 1: Click the Animations tab on the Ribbon.

Step 2: In the Advanced Animation group, click Animation Pane. The Animation Pane opens on the right side of the PowerPoint window.





The Animation Pane displays a list of all the animations added to the selected slide. From this pane, you can rearrange the order of animations, modify their timing, change the start options or remove an animation effect whenever required.

Removing an Animation

If an animation effect is no longer needed, it can be removed easily. Removing an animation does not delete the object from the slide; it only removes the animation effect applied to it. To remove an animation, click the animation number beside the object. When the number box is highlighted, press the Delete key on the keyboard. You can also remove an animation through the Animation Pane. Select the animation effect from the pane and press the Delete key. The selected animation effect is removed from the slide.



Tech Fact

Press Shift + F7 to open the Thesaurus quickly.

A synonym is a word that has the same or nearly same meaning as other word.

Slide Transitions

A presentation is displayed one slide after another. The effect that appears when one slide changes to the next is called a slide transition. Slide transitions make presentations smoother and more attractive by adding movement between slides. PowerPoint provides many transition effects that can be applied to slides. A suitable transition improves the flow of the presentation, while using too many different transitions may distract the audience.

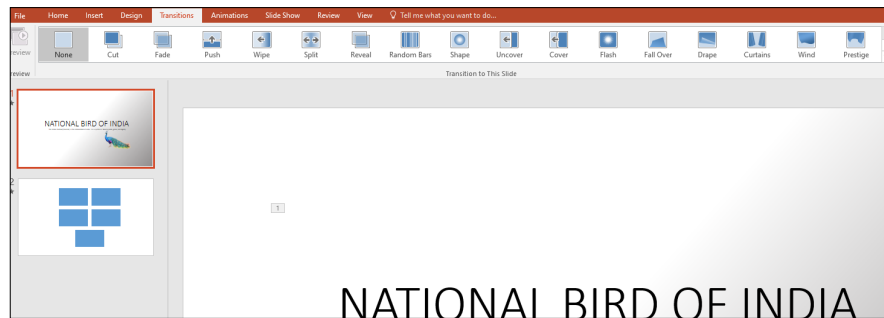
Steps to apply a slide transition are as follows:



Step 1: In the Slide Navigation pane, select the slide on which you want to apply the transition.

Step 2: Click the Transitions tab on the Ribbon.

Step 3: In the Transition to This Slide group, click the More button to display the complete gallery of transition effects.

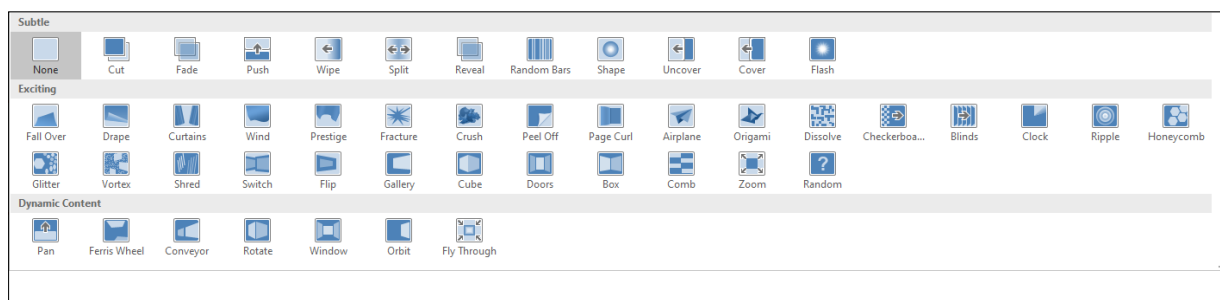


The transition gallery contains three categories of transition effects:

Subtle: These are simple and smooth transition effects suitable for most presentations.

Exciting: These transitions are more colourful and dynamic, making slides appear with stronger visual effects.

Dynamic Content: These transitions mainly affect the contents of the slide, such as text and pictures. If two consecutive slides have the same layout, only the placeholders move instead of the entire slide.



Step 4: Click the transition effect you want to apply. A preview of the selected transition is displayed on the slide.

Timing

After applying a transition, you can control how it behaves during the slide show. PowerPoint allows you to add transition sounds, adjust the duration of the transition and decide whether the next slide should appear on a mouse click or automatically after a specified time.



Steps to Modify Animation Timing

Step 1: Click the Transitions tab.

Step 2: In the Timing group, click the Sound drop-down arrow and select the sound you want to use with the transition.

Step 3: Enter the required time in the Duration box to control how long the transition effect lasts.

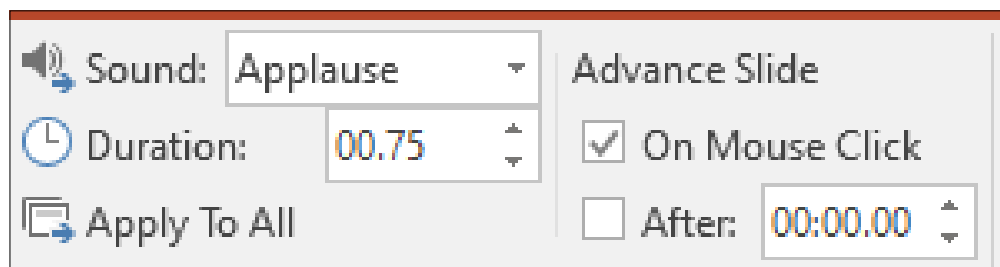
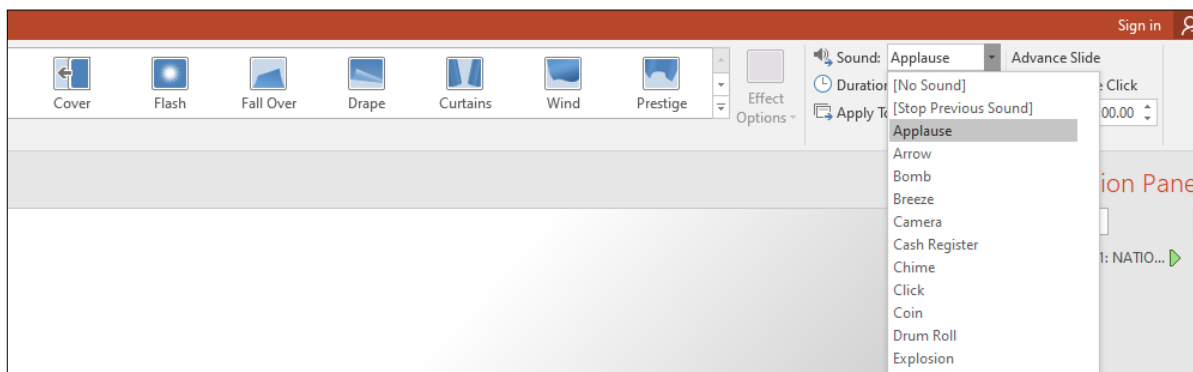
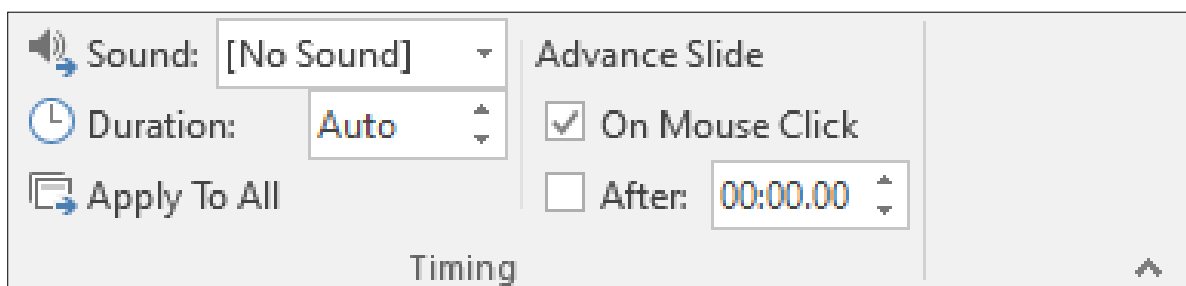
Step 4: Click Apply to All if you want the selected transition and timing settings to be used for every slide in the presentation.

Step 5: In the Advance Slide section, choose how the next slide should appear.

On Mouse Click: The next slide appears only when the mouse is clicked or a key is pressed.

After: The next slide appears automatically after the specified time interval.

These timing options help you control the speed and flow of your presentation, making it easier for the audience to follow each slide.





Activity

Create a PowerPoint presentation on “My Dream School” using 5 slides.

- Apply a suitable theme to the presentation.
- Change the background of at least one slide.
- Insert one SmartArt graphic to present your ideas.
- Add animations to the title and one picture.
- Apply a suitable slide transition to all slides.

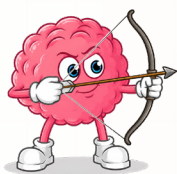
Present your slideshow in class and explain why you chose the theme, animations and transitions.



Quick Recap

- *A theme is a built-in feature that quickly changes the overall design of a presentation using matching colours, fonts and effects.*
- *Theme Variants help you modify the colours, fonts and background styles of a selected theme without changing its basic design.*
- *Slide Master is used to create a common layout and formatting that is automatically applied to all slides in a presentation.*
- *SmartArt is a visual tool that presents information through diagrams, lists and graphics, making ideas easier to understand.*
- *Animations add visual effects to text and objects, making a presentation more attractive and helping to highlight important information.*
- *Slide Transitions create smooth visual effects when moving from one slide to another, making the presentation more engaging.*





TEST YOUR SKILLS



A. Choose the correct option

1. Which tab is used to apply a theme to a presentation?
a. Home ☐ b. Design ☐
c. Review ☐ d. Transition ☐
2. Which option is used to insert a Smart Art graphic?
a. Shapes ☐ b. Smart Art ☐
c. Word Art ☐ d. Image ☐
3. Which animation category is used to make an object appear on a slide?
a. Entrance ☐ b. Exit ☐
c. Emphasis ☐ d. Middle ☐
4. Which feature helps you apply the same layout to all slides?
a. Slide Master ☐ b. Notes Page ☐
c. Zoom ☐ d. Duplicate ☐
5. Which transition category contains simple and basic effects?
a. Dynamic Content ☐ b. Simple ☐
c. Exciting ☐ d. Subtle ☐
6. Which option changes how long a transition effect lasts?
a. Duration ☐ b. Sound ☐
c. Preview ☐ d. Timing ☐

B. Fill in the Blanks

Word Bank: Slide Master, Entrance, Animation Pane, Background, Duration

1. The _____ effect makes an object appear on the slide.
2. The _____ option controls how long an animation/transition runs.
3. You can change colour/picture of a slide by formatting its _____.



4. The _____ lists all animation effects applied to the current slide.
5. _____ is used to maintain a common layout for all slides.

C. State True or False

1. SmartArt can be inserted only on the title slide. _____
2. A theme changes the appearance of a presentation. _____
3. Motion Paths move an object along a selected path. _____
4. The Animation Pane helps you arrange animation effects. _____

D. Answer The Following Questions

1. Why are themes used in a PowerPoint presentation?
2. What is the purpose of Slide Master?
3. Name the four categories of animation effects.
4. How do you insert a SmartArt graphic?
5. What is the difference between animations and slide transitions?

E. Competency-Based Questions

- a. Riya is preparing a presentation on “Healthy Food.” She wants the headings to appear one after another and the slides to change smoothly. Which PowerPoint features should she use? Explain the steps briefly.
- b. Aman has created a presentation with different colours on every slide. He now wants all the slides to have the same design and layout. Which PowerPoint feature will help him? Explain how he can use it.
- c. Aarav has completed his PowerPoint presentation, but the slides look dull and uninteresting. Which formatting features can he use to make the presentation more attractive?

F. Let's Explore

Create a presentation on “My Favourite Festival”.

- Apply a suitable theme.
- Insert one SmartArt graphic.
- Add at least one picture.
- Apply animations to the title.
- Use a slide transition between slides.





Tech Practice

1. Prepare a presentation on any one topic:
 - a. My Favourite Scientist
 - b. Clean India Mission
2. Use WordArt to design an attractive title.



3. Insert suitable pictures and one SmartArt graphic.
4. Apply a theme, change the background of one slide, add animations and use slide transitions.

G. Let's Solve

Guess Who Am I?

I help you present ideas using diagrams instead of plain text. _____

I control how one slide changes to the next. _____

I let you change the appearance of every slide at the same time. _____

I show all the animation effects applied to a slide. _____

Teacher's Note

- Encourage students to compare different themes and explain which one best suits their presentation topic.
- Demonstrate how appropriate use of animations and transitions improves a presentation without making it distracting.



Introduction to Excel 2016

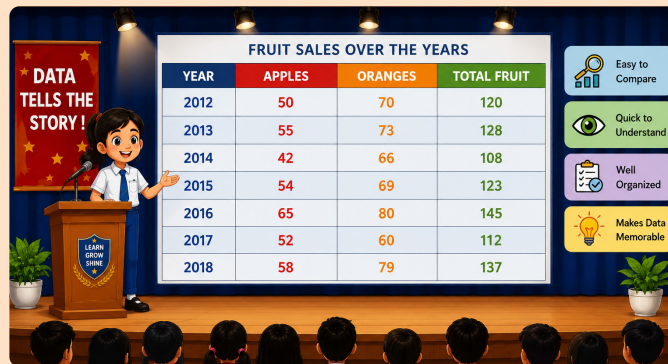


Let's Warm Up the Brain

Learning Goals

- Features of Excel
- Components of Excel
- Changing Active Cell
- Creating New Workbook
- Entering Data
- Working with Worksheets
- Opening a Workbook
- Saving a Workbook
- Closing a Workbook

If you had to present this data in your school assembly, how would you make it interesting and easy to understand?



Excel 2016

Microsoft Excel 2016 is a spreadsheet program used to organise, store and work with different types of data. It allows us to enter text, numbers and formulas in rows and columns. Excel helps us perform calculations quickly and present information in a neat and organised manner. It is widely used in schools, offices and businesses to prepare records, tables and reports.

Features Of Excel 2016

Excel 2016 provides many useful features that make working with data simple and efficient. Some of these features are:

- Excel contains built-in formulas and functions that help us perform calculations quickly and accurately.
- We can format cells, rows and columns using different fonts, colours, borders and styles to make the worksheet attractive.
- Whenever a value is changed, Excel automatically updates the related calculations throughout the worksheet.
- It provides tools such as Sort and Filter to arrange data and find required information easily.



- AutoFill helps us quickly enter a series of numbers, dates or days without typing them repeatedly.
- Excel allows us to create colourful charts and graphs to present data in an easy-to-understand form.

Starting Excel

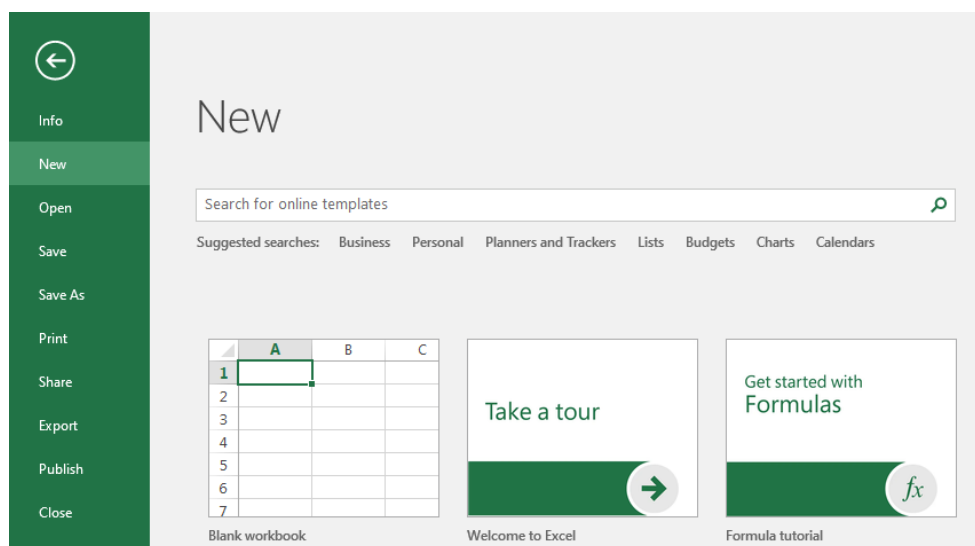
Follow these steps to start Excel 2016.

Step 1: Click on the Start button.

Step 2: Type Excel in the search box or select it from the list of applications.

The Microsoft Excel 2016 window opens on the screen. Click on Blank Workbook to create a new worksheet.

A worksheet is made up of rows and columns where data can be entered, edited and organised. An Excel file is called a workbook. A workbook may contain one or more worksheets. By default, a new workbook opens with a worksheet named Sheet1.



Tech Fact

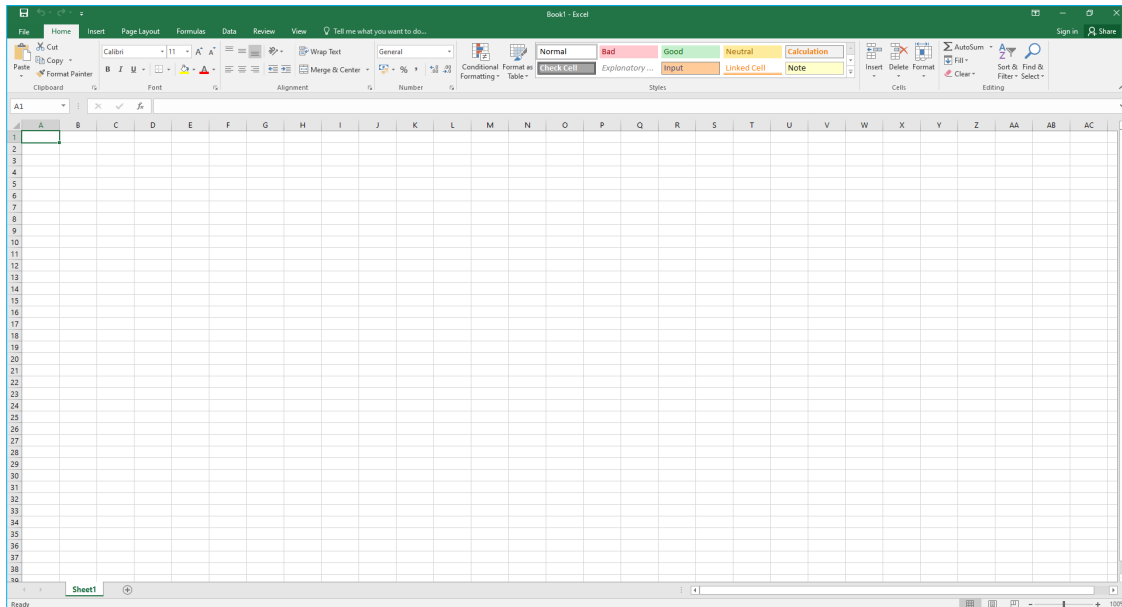
It is used to organize information in rows and columns.

Microsoft Excel 2016 is a spreadsheet program used to organize and calculate data..



Components of Excel 2016 Window

The Excel 2016 window consists of several important parts. Each part has a specific purpose and helps us work more efficiently.



Title Bar

The Title Bar is located at the top of the Excel window. It displays the name of the current workbook along with the name of the application. The Minimize, Maximize/Restore and Close buttons are also available on the right side of the Title Bar.

Quick Access Toolbar

The Quick Access Toolbar contains commonly used commands such as Save, Undo and Redo. It allows us to use these commands with a single click.

Ribbon

The Ribbon is placed below the Title Bar. It contains different tabs such as Home, Insert and Page Layout. Each tab includes groups of related commands and tools.

File Tab

The File Tab is located in the upper-left corner of the Ribbon. It provides options such as New, Open, Save, Save As, Print and Close for managing workbooks.

Status Bar

The Status Bar is found at the bottom of the Excel window. It displays useful information about the current worksheet and selected cells. Messages and calculation results may also appear here.

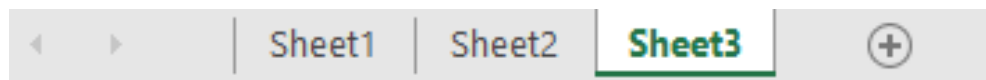


View Bar

The View Bar is located on the right side of the Status Bar. It helps us switch between different worksheet views such as Normal, Page Layout and Page Break Preview.

Sheet Tab

The Sheet Tab is located at the bottom of the Excel window. It displays the names of all the worksheets in a workbook. By default, the first worksheet is named Sheet1. You can switch from one worksheet to another by clicking the required sheet tab.



Rows and Columns

A worksheet is divided into rows and columns. Rows run horizontally and are identified by numbers such as 1, 2, 3.... Columns run vertically and are identified by letters such as A, B, C.... Together, they help organise data in a structured way.

Cells

A cell is formed where a row and a column meet. It is the smallest unit of a worksheet and is used to store text, numbers or formulas. Every cell has a unique address based on its column letter and row number, such as A1 or C5.

Worksheet

A worksheet is the main working area of Excel. It consists of rows, columns and cells where information can be entered and managed. A workbook can contain several worksheets, making it easy to organise different types of data in one file.

Active Cell

The Active Cell is the cell that is currently selected. It is highlighted with a thick border and is ready to accept data. You can type or edit information only in the active cell.

Formula Bar

The Formula Bar is located above the worksheet. It displays the contents of the active cell. You can also use it to enter or edit text, numbers, and formulas without changing the data directly in the cell. It makes editing long formulas and text much easier because you can see the entire content clearly. The Formula Bar updates automatically whenever you select a different cell. It is especially useful for checking and correcting formulas without affecting other cells in the worksheet.



Changing the Active Cell

You can change the active cell by using either the mouse or the keyboard. Simply click on any cell with the mouse, or use the following keys to move the active cell.

Key	Function
Enter Key	Moves the active cell one row down.
Shift + Enter Key	Moves the active cell one row up.
Shift + Tab Key	Moves the active cell one column to the left.
Tab Key	Moves the active cell one column to the right.
[Left Arrow Key]	Moves the active cell one column to the left.
[Right Arrow Key]	Moves the active cell one column to the right.
[Up Arrow Key]	Moves the active cell one row up.
[Down Arrow Key]	Moves the active cell one row down.
Ctrl + Home Key	Moves the active cell to the first cell (A1).
Ctrl + End Key	Moves the active cell to the last used cell in the worksheet.

Creating a New Workbook

A workbook is an Excel file that contains one or more worksheets. Whenever you want to create a new spreadsheet, you need to create a new workbook first.

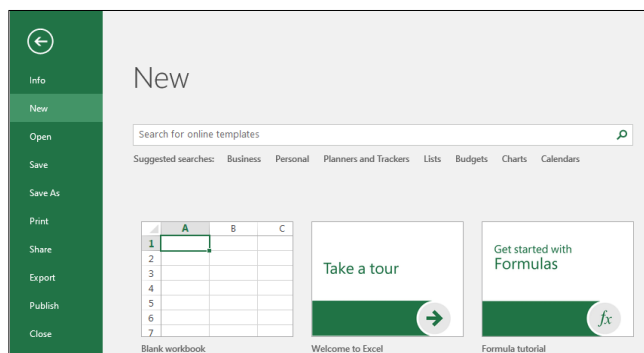
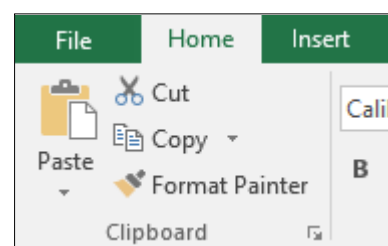
Follow these steps to create a new workbook.

Step 1: Open Microsoft Excel 2016.

Step 2: Click on the File tab.

Step 3: Select the New option from the left pane.

Step 4: Click on Blank Workbook.



Entering Data

Data can be entered into a worksheet by typing in the active cell.

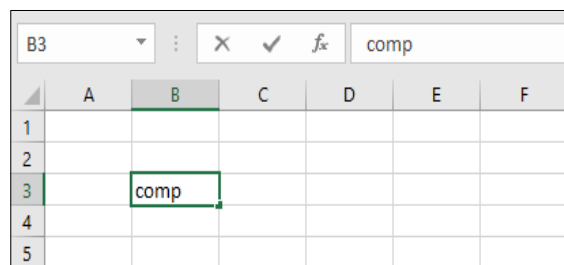
Follow these steps to enter data.

Step 1: Click on the cell where you want to enter the data.

Step 2: Type the required text, number or date.

Step 3: Press the Enter key.

The entered data appears in the selected cell.



Working with Worksheet

A workbook can contain several worksheets. You can move between worksheets, rename them, add new ones or remove unwanted worksheets whenever required.

Renaming a Worksheet

To rename a worksheet, follow these steps.

Method 1

Step 1: Right-click on the sheet tab that you want to rename.

Step 2: Select the Rename option.

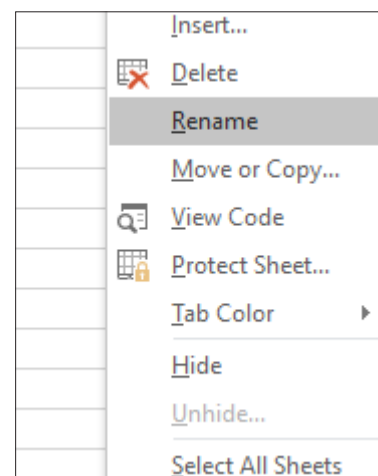
Step 3: Type the new worksheet name and press the Enter key.

Method 2

Step 1: Double-click on the sheet tab.

Step 2: Type the new name and press the Enter key.

The worksheet is renamed.



Adding a New Worksheet

To insert another worksheet into a workbook, follow these steps.

Method 1

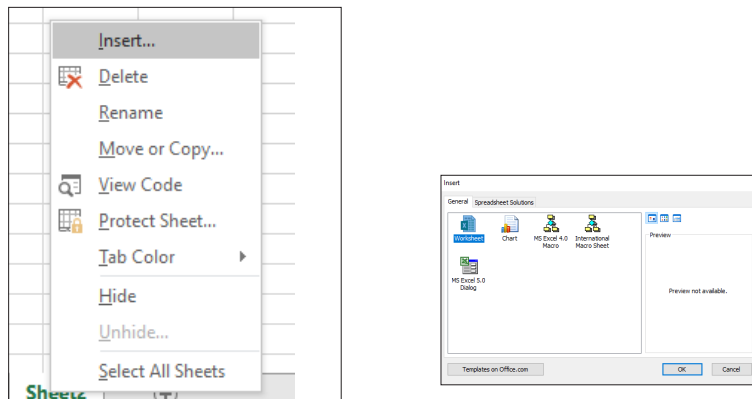
Step 1: Right-click on any sheet tab.

Step 2: Select the Insert option.

Step 3: Choose Worksheet and click OK.

Method 2

Click on the New Sheet (+) button next to the sheets tabs, a new worksheet is created.



Removing a Worksheet

To remove a worksheet, follow these steps.

Method 1

Step 1: Right-click on the sheet tab that you want to remove.

Step 2: Select the Delete option.

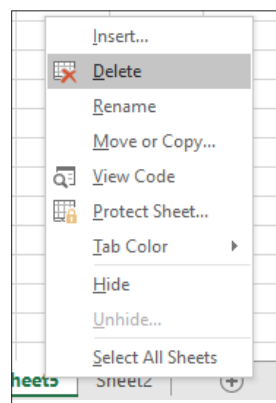
Method 2

Step 1: Click on the Home tab.

Step 2: In the Cells group, click the Delete drop-down arrow.

Step 3: Select Delete Sheet.

The worksheet is deleted from the workbook.



Saving a Workbook

To save a workbook, follow these steps.

Step 1: Click on the File tab.

Step 2: Click Save if you are saving the workbook for the first time. To save the workbook with a different name, click Save As.

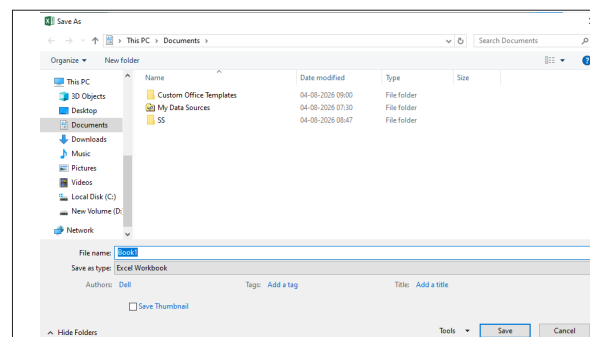
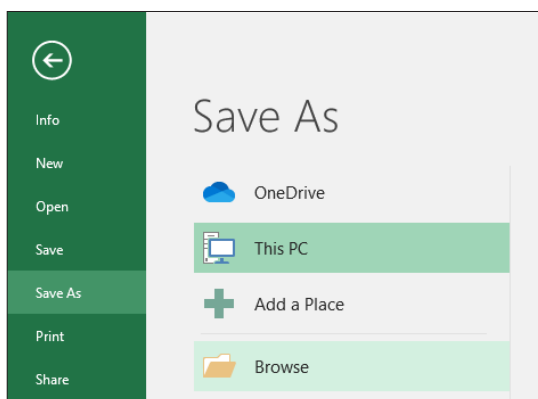
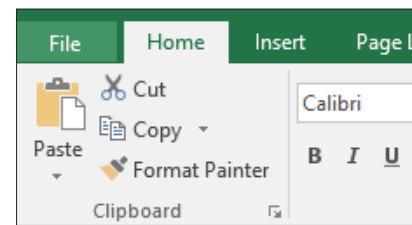
The Save As window appears.

Step 3: Click on Browse and choose the folder where you want to save the workbook.

Step 4: Type a suitable name in the File name box.

Step 5: Click the Save button.

The workbook is saved in the selected location.



Opening a Workbook

To open a workbook, follow these steps.

Step 1: Click on the File tab.

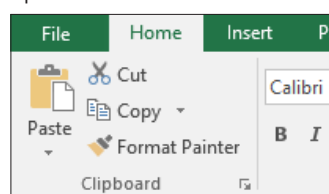
Step 2: Click Open option from the left pane

Step 3: Click on Browse, the Open dialog box will appear.

Step 4: Go to the folder where your workbook is saved and select the required file.

Step 5: Click the Open button.

The selected workbook will be opened.



Closing a Workbook

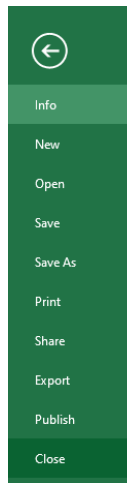
You can close a workbook after finishing your work without closing the Excel application.

Follow these steps.

Step 1: Click on the File tab.

Step 2: Select the Close option.

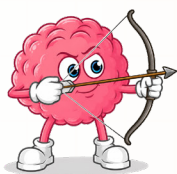
The current workbook closes, while Microsoft Excel remains open.



Quick Recap

- *Microsoft Excel 2016 is an electronic spreadsheet program used to organise and manage data in rows and columns.*
- *A workbook consists of one or more worksheets where data can be entered, edited and stored.*
- *Excel provides built-in formulas, functions and formatting tools to make calculations and present data effectively.*
- *The Excel window contains components such as the Title Bar, Ribbon, Formula Bar, Name Box, Status Bar, Sheet Tabs and Worksheet area.*
- *An active cell is the selected cell where data is entered or edited.*
- *Workbooks can be created, saved, opened, renamed and managed using the commands available in the File tab.*
- *Worksheets can be added, renamed or deleted whenever required.*





TEST YOUR SKILLS



A. Choose the correct option

1. Which feature helps Excel perform calculations automatically?

- | | | | |
|-------------|-----|--------------|-----|
| a. Themes | () | b. Functions | () |
| c. Pictures | () | d. Shapes | () |

2. Which key moves the active cell one column to the left?

- | | | | |
|------------------|-----|-----------------|-----|
| a. Tab | () | b. Enter | () |
| c. Shift + Enter | () | d. Ctrl + Enter | () |

3. Which part of Excel displays the contents of the active cell?

- | | | | |
|----------------|-----|---------------|-----|
| a. Ribbon | () | b. Name Box | () |
| c. Formula Bar | () | d. Status Bar | () |

4. Which option is used to create a new workbook?

- | | | | |
|---------|-----|----------|-----|
| a. Save | () | b. Print | () |
| c. Open | () | d. New | () |

B. Fill in the Blanks

Word Bank: Workbook, Ctrl + S, Worksheet, Active Cell

1. A spreadsheet page inside a workbook is called a _____.
2. The selected cell where data is entered is known as the _____.
3. _____ is the shortcut key used to save a workbook. _____.
4. An Excel file is called a _____.

C. State True or False

1. Excel stores data in rows and columns. _____
2. A workbook can contain only one worksheet. _____
3. The Formula Bar is used to enter or edit data. _____
4. The Sheet Tab is located at the bottom of the Excel window. _____



D. Answer The Following Questions

1. Write any two features of Excel 2016.
2. Write the steps to save a workbook.
3. What is a workbook? How do you create a new workbook?
4. What is an active cell?

E. Competency-Based Questions

- a. Rahul has entered the marks of his classmates in Excel. He wants to keep the information neat and easy to read. Which Excel features or components will help him organise the data properly? Explain.
- b. Meera has completed her science project in Excel. She wants to save it in a folder named School Projects with the file name Science Project. Describe the steps she should follow to save her workbook correctly.

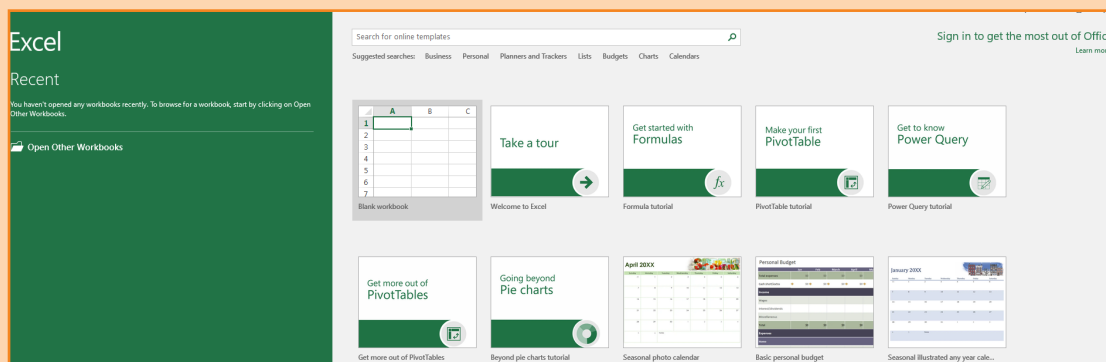
F. Let's Get Better

1. Your friend accidentally deleted a worksheet. Which method can you suggest to create a new worksheet again?
2. You have finished entering data in Excel. Explain how you will save your workbook before closing it.
3. Your class teacher asks you to prepare a student record in Excel. What headings would you include in your worksheet?



Tech Practice

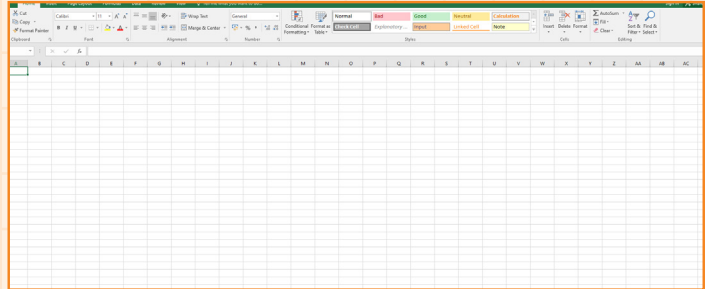
Create the following worksheet and save it with your name.



G. Let's Solve

See the given table and write cell addresses of the following:

Maths _____
English _____
Science _____
Computer _____
Art _____
Music _____
EVS _____
GK _____



Teacher's Note

- Encourage students to identify different parts of the Excel window.
- Demonstrate how to move between cells using both the mouse and keyboard.
- Give students simple worksheets to enter and organise data neatly



Editing Cell Contents in Excel 2016



Let's Warm Up the Brain

Learning Goals

- Selecting Cells
- Entering Date & Time
- Changing Cell Content
- Undo & Redo Commands
- Copying & Moving Data
- Deleting Cell Contents
- Using Auto Fill

Why does every cell in a worksheet have a unique address?

Why is it helpful to store data in rows and columns? Justify your answer.

If one more student is added to the list, where will the new record be entered?

Selecting Cells

Before editing any data in a worksheet, you need to select the required cell or cells. The selected cell becomes the active cell, allowing you to enter, edit, copy, or delete data. Let us learn the different ways to select cells in Excel.

Selecting a Cell

To select a single cell, follow these steps:

Step 1: Click on the cell you want to select.

Step 2: The selected cell becomes the active cell and is surrounded by a thick border. You can also use the arrow keys to move to another cell.

Selecting a Row

To select a row, follow these steps:

Step 1: Move the mouse pointer to the row number of the row you want to select. The pointer changes into a right-pointing arrow.

Step 2: Click on the row number. The complete row gets selected.



	A	B	C	D	E	F
1						
2						
3			NAME	AGE	CLASS	
4			HARSH	11	4	
5			NAMAN	11	4	
6			ARYAN	12	5	
7			KRISH	11	4	
8			VAIBHAV	12	5	
9			RAHUL	11	4	
10			VISHAL	11	4	

Selecting a Column

To select a column, follow these steps:

Step 1: Move the mouse pointer to the column heading of the column you want to select. The pointer changes into a down-pointing arrow.

Step 2: Click on the column heading. The entire column gets selected.

A	B	C	D	E	F
		NAME	AGE	CLASS	
		HARSH	11	4	
		NAMAN	11	4	
		ARYAN	12	5	
		KRISH	11	4	
		VAIBHAV	12	5	

Selecting a Range of Cells

A range is a group of two or more adjacent cells that form a rectangular area in a worksheet. A range is identified by the addresses of the first and the last cell, separated by a colon (:). For example, the range B2:D6 begins at cell B2 and ends at cell D6. Selecting a range allows you to perform the same operation on all the selected cells at once, such as formatting, copying, deleting, or entering data.

			NAME	AGE	CLASS	
			HARSH	11	4	
			NAMAN	11	4	
			ARYAN	12	5	
			KRISH	11	4	
			VAIBHAV	12	5	
			RAHUL	11	4	
			VISHAL	11	4	

Using the Mouse

To select a range of cells using the mouse, follow these steps:

Step 1: Click on the first cell of the required range.

Step 2: Keep the left mouse button pressed and drag the pointer across the cells you want to include.

Step 3: Release the mouse button after reaching the last cell. All the selected cells become highlighted, while the first cell remains the active cell.

Using the Keyboard

You can also select a range of cells by using the keyboard.

Step 1: Select the first cell of the required range.

Step 2: Press and hold the Shift key.

Step 3: Press the Arrow keys to extend the selection in the desired direction.

Step 4: Release the Shift key after selecting the required range.

Selecting Multiple Cells

To select cells that are not next to each other, click on the first cell. Now, press and hold the Ctrl key and click on the other cells you want to select. All the selected cells are highlighted.

NAME	AGE	CLASS
HARSH	11	4
NAMAN	11	4
ARYAN	12	5
KRISH	11	4
VAIBHAV	12	5
RAHUL	11	4

Selecting the Whole Worksheet

To select the entire worksheet, click the Select All button located at the top-left corner of the worksheet, above the row numbers and before column A.

	A	B	C	D	E	F
1						
2						
3			NAME	AGE	CLASS	
4			HARSH	11	4	
5			NAMAN	11	4	
6			ARYAN	12	5	
7			KRISH	11	4	
8			VAIBHAV	12	5	
9			RAHUL	11	4	
10			VISHAL	11	4	
11			ARMAAN	12	5	
12						



Entering Date & Time

Excel can recognise dates and times entered in different formats. It automatically stores them as date or time values and displays them in a suitable format. You can also change the display format whenever required.

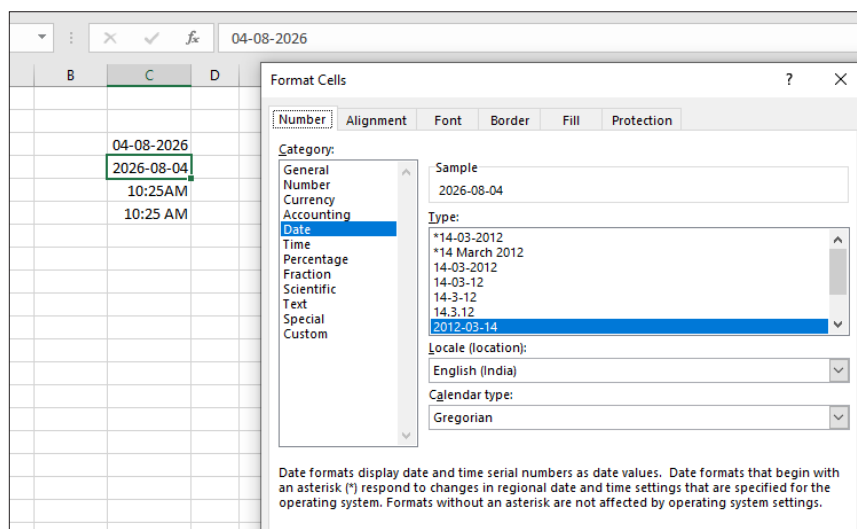
To change the format of a date or time, follow these steps:

Step 1: Select the cell containing the date or time.

Step 2: Click on the Home tab.

Step 3: Click the Number Format dialog box launcher.

Step 4: In the Format Cells dialog box, choose Date or Time from the Category list.



Step 5: Select the format you want to apply.

Step 6: Click the OK button.



Tech Fact

Ctrl + ; Inserts the current date.

Ctrl + Shift + ; Inserts the current time.

Changing Cell Contents

The information entered in a cell can be changed whenever required. You can edit the contents of a cell without deleting the existing worksheet.

Cell contents can be edited in two ways: **Replacing Cell Content & Using the Formula Bar.**



Replacing Cell Contents

If you enter new data in a selected cell, the old contents are replaced automatically. To replace the contents of a cell, follow these steps:

Step 1: Click on the cell whose contents you want to replace.

Step 2: Type the new data.

Step 3: Press the Enter key. The new data replaces the previous contents.

Using the Formula Bar

The Formula Bar is located above the worksheet. It displays the contents of the active cell and allows you to view or edit the data without changing it directly in the cell. It is especially useful when a cell contains long text, numbers, or formulas, as the complete contents can be seen clearly in the Formula Bar.

To edit the contents of a cell using the Formula Bar, follow these steps:

Step 1: Select the cell whose contents you want to edit.

Step 2: Click inside the Formula Bar. The contents of the selected cell appear in it.

Step 3: Make the required changes in the Formula Bar.

Step 4: Press the Enter key or click the Enter button on the Formula Bar to save the changes.

The updated contents are displayed in the selected cell.

A	B	C	D	E
		NAME	AGE	CLASS
		HARSH	11	4
		NAMAN	11	4
		ARYAN	12	5
		KRISH	11	4
		VAIBHAV	12	5
		RAHUL	11	4
		VISHAL	11	4
		ARMAAN	12	5

Undo and Redo Commands

While working in a worksheet, you may sometimes enter incorrect data or perform an action by mistake. Microsoft Excel provides the Undo command to reverse the last action performed. You can use the Undo command several times to cancel a series of recent actions. The Undo button is available on the Quick Access Toolbar, and you can also use the shortcut key Ctrl + Z.

The Redo command is used to restore an action that has been undone. If you undo a change by mistake, you can use the Redo command to bring it back. The Redo button is also available on the Quick Access Toolbar, and the shortcut key for Redo is Ctrl + Y.

These commands help you correct mistakes quickly and make your work easier.



Copy and Move Data

Excel allows you to copy or move data from one cell to another. Copying creates a duplicate of the selected data, whereas moving transfers the data from one location to another.

Copy Data

To copy data from one cell to another, follow these steps:

Step 1: Select the cell or range of cells that you want to copy.

Step 2: Click the Copy button on the Home tab or press Ctrl + C.

Step 3: Select the destination cell.

Step 4: Click the Paste button or press Ctrl + V.

The copied data appears in the selected location, while the original data remains unchanged.

	NAME	AGE	CLASS
	HARSH	11	4
	NAMAN	11	4
	ARYAN	12	5
	KRISH	11	4
	VAIBHAV	12	5
	RAHUL	11	4
	VISHAL	11	4
	ARMAAN	12	5

[illegible]

Move Data

Moving data transfers the selected data from one place to another. The data is removed from its original location and placed in the new location.

To move data, follow these steps:

Step 1: Select the cell or range of cells that you want to move.

Step 2: Click the Cut button on the Home tab or press Ctrl + X.

Step 3: Select the destination cell.

Step 4: Click the Paste button or press Ctrl + V.

The selected data is moved to the new location.

	NAME	AGE	CLASS
	HARSH	11	4
	NAMAN	11	4
	ARYAN	12	5
	KRISH	11	4
	VAIBHAV	12	5
	RAHUL	11	4
	VISHAL	11	4
	ARMAAN	12	5

	NAME	AGE	CLASS
	HARSH	11	4
	NAMAN	11	4
	VAIBHAV	12	5
	RAHUL	11	4
	VISHAL	11	4
	ARMAAN	12	5
	ARYAN	12	5
	KRISH	11	4

Deleting Cell Contents

Sometimes, you may need to remove the contents of a cell while keeping the cell itself. Excel allows you to clear only the data without affecting the worksheet structure.

Deleting the Cells

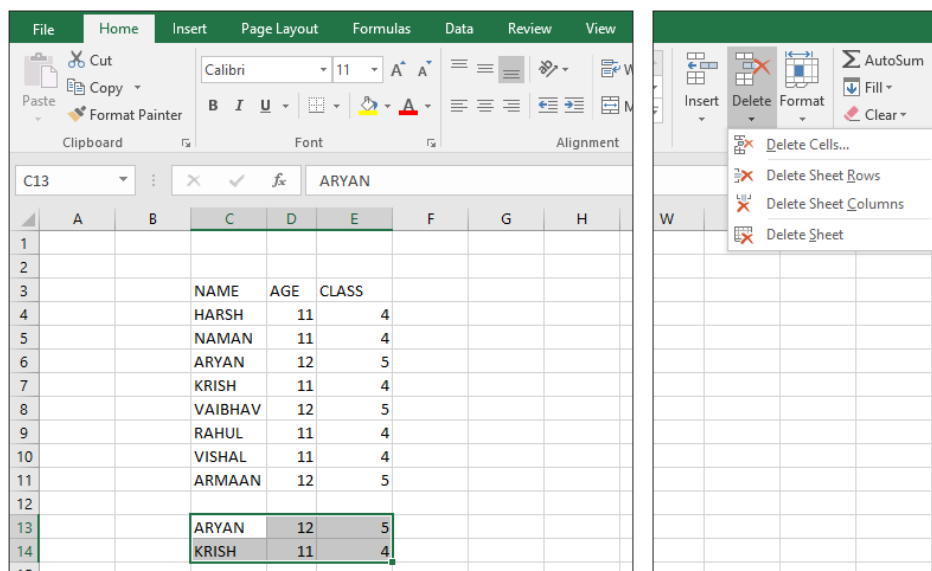
To delete the contents of a cell, follow these steps:

Step 1: Select the cell you want to delete.

Step 2: Click the Delete drop-down arrow in the Cells group on the Home tab.

Step 3: Click Delete Cells.

Step 4: In the Delete dialog box, choose the required option and click OK.



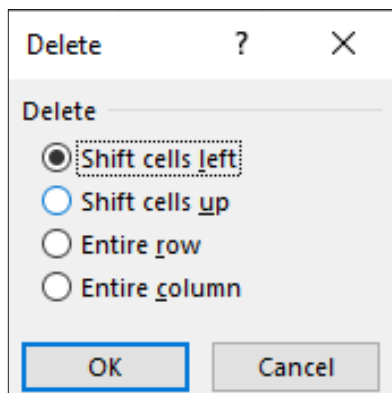
The Delete Dialog box offers the following options:

Shift cells left: Moves the cells on the right towards the left.

Shift cells up: Moves the cells below upward.

Entire row: Deletes the complete row containing the selected cell.

Entire column: Deletes the complete column containing the selected cell.



3			NAME	AGE	CLASS	
4			HARSH	11	4	
5			NAMAN	11	4	
6			ARYAN	12	5	
7			KRISH	11	4	
8			VAIBHAV	12	5	
9			RAHUL	11	4	
10			VISHAL	11	4	
11			ARMAAN	12	5	
12						
13						
14						

Deleting an Entire Row or Column

You can also remove a complete row or column if it is no longer required.

To delete a row or column, follow these steps:

Step 1: Select the required row or column.

Step 2: Click the Delete drop-down arrow in the Cells group on the Home tab.

Step 3: Choose Delete Sheet Rows or Delete Sheet Columns.

	A	B	C	D	E	F	G	H	I
1									
2									
3			NAME	AGE	CLASS				
4			HARSH	11	4				
5			NAMAN	11	4				
6			ARYAN	12	5				
7			KRISH	11	4				
8			VAIBHAV	12	5				



Using Auto Fill Feature

The Auto Fill feature helps you fill a series of data automatically in a worksheet. It can be used to copy the same value or continue a sequence of numbers, dates, days, months, and other recognised patterns. The Fill Handle, a small square at the bottom-right corner of the selected cell, is used to perform Auto Fill.

To use the Auto Fill feature, follow these steps:

Step 1: Enter the required data in a cell. If you want to create a sequence, enter the first value or the first few values of the series.

Step 2: Select the cell or range of cells containing the data.

Step 3: Move the mouse pointer to the Fill Handle located at the bottom-right corner of the selected cell. The pointer changes into a small black plus (+) sign.

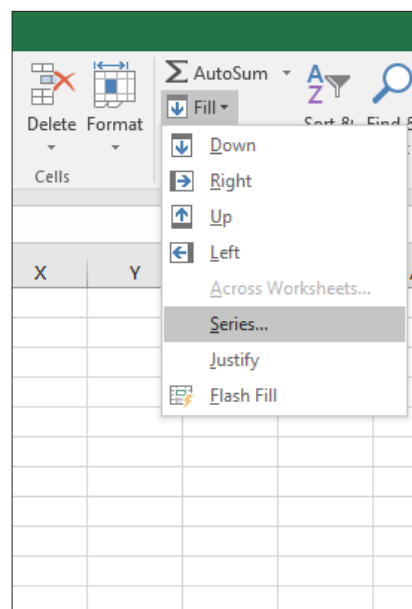
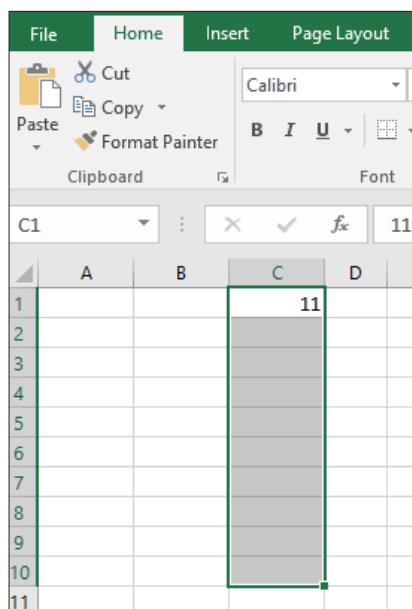
Step 4: Press and hold the left mouse button, then drag the Fill Handle over the cells where you want to fill the data.

Step 5: Release the mouse button after selecting the required cells. Excel automatically copies the data or continues the series according to the recognised pattern.

Step 6: If the Auto Fill Options button appears, click it to choose the required filling option, such as copying cells, filling a series, or formatting only.

	A
1	11
2	12
3	13
4	14
5	15
6	
7	
8	
9	
10	
11	

	A
1	11
2	12
3	13
4	14
5	15
6	16
7	17
8	18
9	19
10	20
11	



Alternate Method

Step 1: Enter the required data in the first cell or enter the starting values of the series.

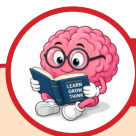
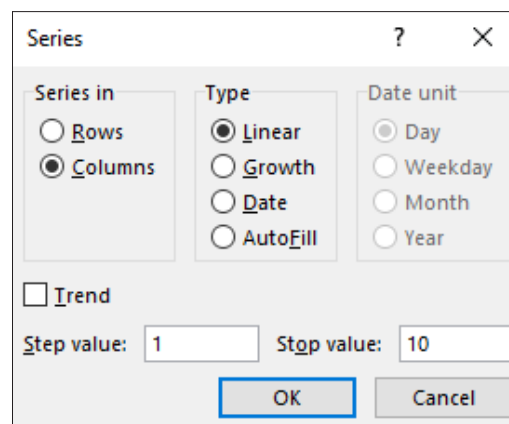
Step 2: Select the cell containing the data along with the cells that you want to fill.

Step 3: Click the Home tab. In the Editing group, click the Fill drop-down arrow.

Step 4: Select Series from the list. The Series dialog box appears.

Step 5: Choose the required options, such as Rows or Columns, select the Type of series, and enter the Step value and Stop value, if required.

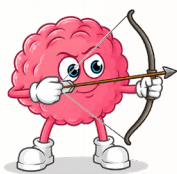
Step 6: Click the OK button. Excel fills the selected cells according to the specified series.



Quick Recap

- A cell or a group of cells must be selected before performing any editing task.
- Rows, columns, ranges, and the entire worksheet can be selected in different ways.
- Cell contents can be edited directly in the cell or through the Formula Bar.
- The Undo and Redo commands help correct or restore actions.
- Copy creates another copy of the data, while Move transfers the data to a new location.
- The Delete key removes the contents of selected cells without deleting the cells.
- Entire rows, columns, or individual cells can also be deleted when required.
- Auto Fill helps enter repeated or sequential data quickly and accurately.





TEST YOUR SKILLS



A. Choose the correct option

- Which part of a worksheet shows the selected cell?
a. Formula Bar ☐ b. Active Cell ☐
c. Status Bar ☐ d. Ribbon ☐
- Which of the following is a correct range?
a. A2;D8 ☐ b. C5-D10 ☐
c. B3:E7 ☐ d. 4A:6C ☐
- Which key helps you edit the contents of the active cell?
a. Esc ☐ b. Tab ☐
c. Shift ☐ d. F2 ☐
- Which shortcut key is used to move selected data?
a. Ctrl + C ☐ b. Ctrl + X ☐
c. Ctrl + Z ☐ d. Ctrl + A ☐

B. Fill in the Blanks

Word Bank: Workbook, Ctrl + S, Worksheet, Active Cell

- The selected cell is called the _____.
- The _____ is used to drag and continue a series.
- Press the _____ key to remove the contents of a cell.
- The process of changing the contents of a cell is called _____.
- Hold the _____ key while selecting non-adjacent cells.

C. State True or False

- A worksheet can contain only one cell. _____
- You can edit a cell using the Formula Bar. _____
- The Undo command restores the last undone action. _____



4. The Auto Fill feature can continue number series. _____
5. Deleting cell contents also deletes the cell. _____

D. Answer The Following Questions

1. What is the purpose of the Undo command?
2. What is a range? How can you select a range of cells using the keyboard?
3. Write the steps to move data from one location to another.
4. What is the Fill Handle? How is it used in Auto Fill?
5. Write the steps to enter the current date in a worksheet.

E. Competency-Based Questions

- a. Riya has prepared a list of sports items in one worksheet. She wants to move the complete list to another worksheet and then make a copy of only the item names into a new column. Explain how she can complete both tasks.
- b. Aman has typed 5 and 10 in two consecutive cells. He wants Excel to continue the same pattern up to 50. Which feature should he use? Write the steps.

F. Let's Explore

Find out two more useful features available in the Home tab of Excel. Write their names and discuss their uses with your teacher or classmates.

Always wait patiently for your turn while using the computer lab. Help your classmates whenever they need assistance in using Excel.

G. Let's Solve

Write the shortcut keys for the following:

- | | |
|--|-------|
| Copy the selected cell(s). | _____ |
| Cut (move) the selected cell(s). | _____ |
| Paste the copied or cut data. | _____ |
| Undo the last action. | _____ |
| Redo the last undone action. | _____ |
| Select the entire worksheet. | _____ |
| Edit the contents of the active cell. | _____ |
| Delete contents of the selected cells. | _____ |





Tech Practice

Use the Auto Fill feature and complete the following tasks:

- Fill serial numbers from 101 to 110 in Column A.
- Fill the names of the 12 months in Column C.
- Fill the days of the week in Column E.
- Delete Column B and Column D.

	A	B	C	D	E
1	Sr No.	Items	Months	Info	Day
2	101	Item 1	January	Info 1	Sunday
3	102	Item 2	February	Info 2	Monday
4	103	Item 3	March	Info 3	Tuesday
5	104	Item 4	April	Info 4	Wednesday
6	105	Item 5	May	Info 5	Thursday
7	106	Item 6	June	Info 6	Friday
8	107	Item 7	July	Info 7	Saturday
9	108	Item 8	August	Info 8	Sunday
10	109	Item 9	September	Info 9	Monday
11	110	Item 10	October	Info 10	Tuesday

Teacher's Note

- Encourage students to practise selecting, editing, copying, moving, and deleting data in Excel.
- Demonstrate the use of the Formula Bar, Undo, Redo, and Auto Fill with classroom examples.
- Allow students to perform the chapter activities individually to improve their spreadsheet skills.



Internet and E-Mail

Learning Goals

- Internet
- Requirements to connect to Internet
- Types of connections
- Using Web Browser
- Using URLs
- E-Mail



Let's Warm Up the Brain

Tick the correct uses of Internet.

- | | | | |
|--------------------|-----|---------------|-----|
| Search information | () | Find a recipe | () |
| Take a bath | () | Sending mails | () |
| Download an image | () | Do shopping | () |
| Watch Movies | () | Pack your bag | () |

The Internet is one of the most useful inventions of modern times. It connects millions of computers across the world and allows people to share information and communicate with one another. We can use the Internet to search for information, watch videos, play games, shop online, and send messages.

Internet

The Internet is a global network of millions of computers and computer networks connected all over the world. It allows people to access, share, and exchange information quickly. The Internet contains billions of websites that provide information on different topics. Every website is made up of one or more web pages. Each website has a unique address called a Uniform Resource Locator (URL), which helps users open the required website.

Requirements to connect to Internet

To access the Internet, a computer or other digital device requires some basic hardware and software. These components help establish a connection and allow users to browse websites.

Computer System

A computer system, laptop, tablet, or smartphone is needed to access the Internet. It is used to run applications, process information, and display web pages.



Telephone and Cable Lines

Telephone or cable lines help establish a connection between the user's device and the Internet service. These lines transfer data from one place to another.



Modem

A Modem (Modulator-Demodulator) is a device that converts digital data into signals and receives signals from the Internet. It enables communication between the computer and the Internet service provider.



Web Browser

A Web Browser is application software used to open and view websites. It allows users to search for information and access web pages on the Internet. Some popular web browsers are Google Chrome, Microsoft Edge, Mozilla Firefox, Opera, and Safari.

ISP

An Internet Service Provider (ISP) is a company that provides Internet services to users for a monthly or yearly charge. Some popular ISPs in India are Jio, Airtel, BSNL, and Vi (Vodafone Idea).



Types of Internet Connection

Different types of Internet connections are available to suit different users and devices.

Dial-Up Connection

A Dial-up Connection connects to the Internet through a telephone line. It is one of the oldest types of Internet connections. It is slower than modern Internet connections and cannot be used for phone calls while connected.

Broadband Connection

A Broadband Connection is a high-speed Internet connection that provides fast and reliable access to the Internet. It uses telephone lines, cable TV lines, or optical fibre cables to transmit data at high speed. Broadband allows users to browse websites, download files, watch online videos, play games, and attend online classes without interruption. Unlike a dial-up connection, broadband provides an always-on Internet connection, so users do not need to connect every time they want to access the Internet.

Wi-Fi

Wi-Fi (Wireless Fidelity) is a wireless technology that allows computers, laptops, tablets, smartphones, and other digital devices to connect to the Internet without using cables. It uses radio waves to transmit data between the device and a wireless



router. A Wi-Fi connection provides high-speed Internet access within a limited range, such as in a home, school, office, or public place.



Mobile Internet

Mobile Internet allows users to access the Internet through smartphones or tablets using a mobile network such as 4G or 5G. It enables users to browse the Internet from almost anywhere with network coverage.

Hotspots

A Hotspot is a place or device that provides wireless Internet access using Wi-Fi technology. It allows nearby devices to connect to the Internet without using cables. Hotspots are commonly available in schools, airports, railway stations, hotels, and public places.

Using Web Browser

A Web Browser is an application software that helps us access websites and search for information on the Internet. Before opening a web browser, make sure your computer is connected to the Internet. Popular web browsers include Google Chrome, Microsoft Edge, Mozilla Firefox, Opera, and Safari.

Steps to Open a Web Browser

Step 1: Click on the Start button on the taskbar.

Step 2: From the list of applications, click on the name of your web browser.

Step 3: The web browser window opens, and you are ready to browse the Internet.

Using URLs

Every website has a unique address called a Uniform Resource Locator (URL). A URL helps the web browser locate and open a particular website. When you type a URL correctly, the required web page opens on your screen.

Steps to Open a Website Using a URL

Step 1: Open a web browser.

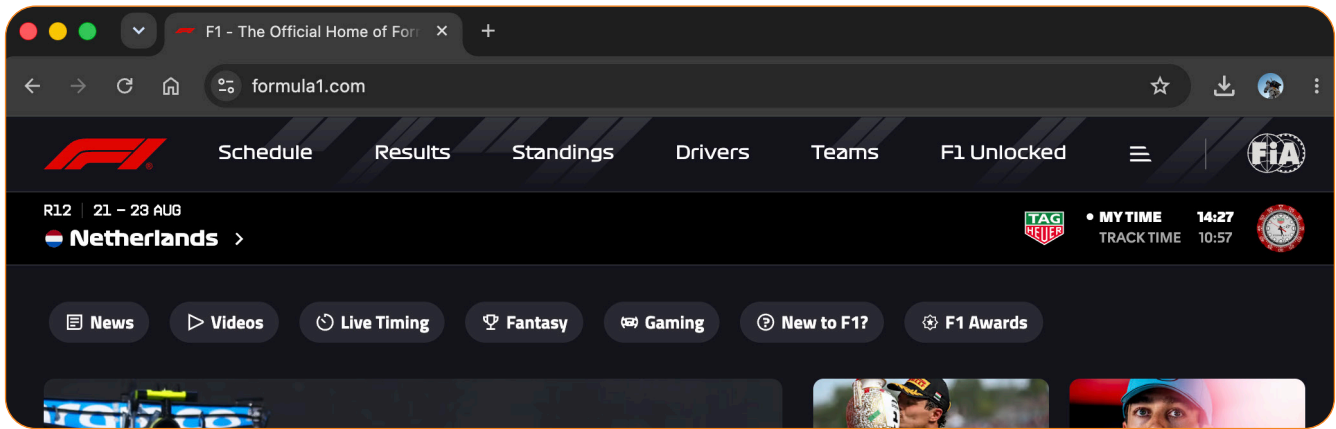
Step 2: Click in the Address Bar.

Step 3: Type the website address (URL).



Step 4: Press the Enter key.

Step 5: The required website opens in the browser window.



Tech Fact

A Hyperlink is a piece of text, image, or button connected to another web page or website. Clicking on a hyperlink takes you directly to the linked page.

E-Mail

Electronic Mail (E-mail) is a method of sending and receiving messages electronically through the Internet. It allows people to communicate quickly with anyone around the world. Besides text messages, users can also send pictures, documents, videos, and other files through e-mail. An Internet connection is required to send and receive e-mails. Many schools, offices, and organisations use e-mail for fast and secure communication.

Advantages of E-Mail

E-mail offers many benefits over traditional postal mail. It helps people communicate quickly and efficiently.

- An e-mail can be sent and received anywhere in the world.
- It reaches the receiver within a few seconds.
- Pictures, documents, videos, and other files can be attached with an e-mail.
- One e-mail can be sent to many people at the same time.
- It is faster and more economical than sending letters by post.
- Messages can be stored and accessed whenever required.



E-Mail Address

Every user needs an e-mail address to send and receive e-mails. An e-mail address is unique and identifies a user on the Internet.

Example: orangeeducation123@gmail.com

orangeeducation123 – Username

@ – At symbol

gmail.com – Domain name

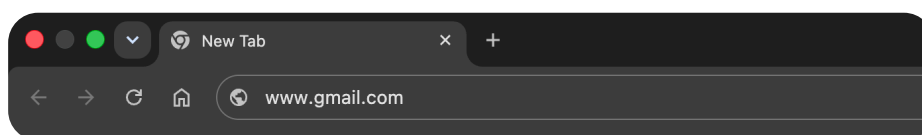
The username identifies the user, while the domain name identifies the e-mail service provider. No two users can have the same e-mail address on the same domain.

Creating an E-Mail Address

To send and receive e-mails, you must first create an e-mail account. Many websites such as Gmail, Outlook, and Yahoo Mail provide free e-mail services. In this chapter, we will learn how to create a Gmail account.

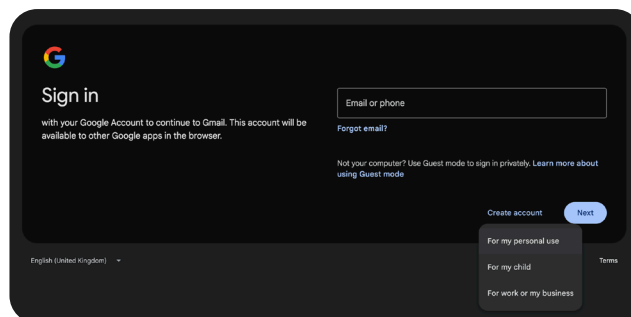
Steps to Create a Gmail Account

Step 1: Open a web browser and type www.gmail.com in the Address Bar. Press the Enter key.



Step 2: Click on the Create account option.

Step 3: Choose For my personal use or For work or my business, as required.



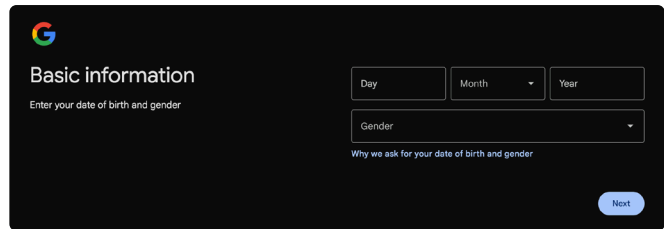
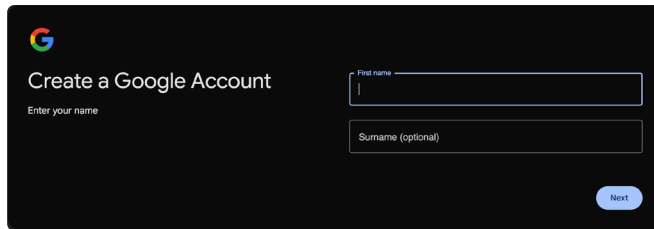
Step 4: Enter your First Name and Last Name, then click Next.

Step 5: Enter your Date of Birth and Gender, then click Next.

Step 6: Choose one of the suggested Gmail addresses or create your own unique

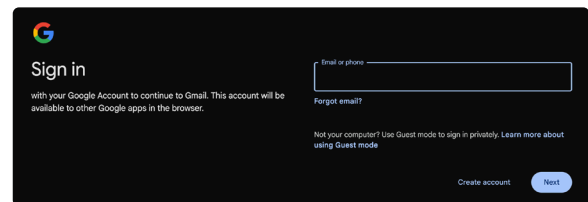
e-mail address. Click Next.

Step 7: Create a strong password, confirm it, and click Next.

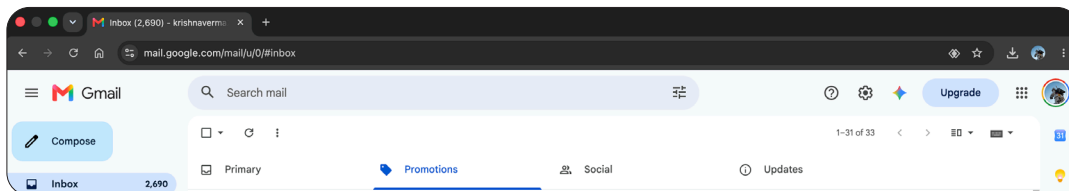


Step 8: Enter your mobile number (or your parent's mobile number with permission) and click Next.

Step 9: Enter the verification code received on the mobile number. Add a recovery e-mail address if required, review your details, and click Next.



Step 10: Read the Privacy Policy and Terms carefully, then click I Agree. Your Gmail account is now ready to use.



Signing In

After creating an e-mail account, you can sign in anytime to read or send e-mails.

Step 1: Open a web browser.

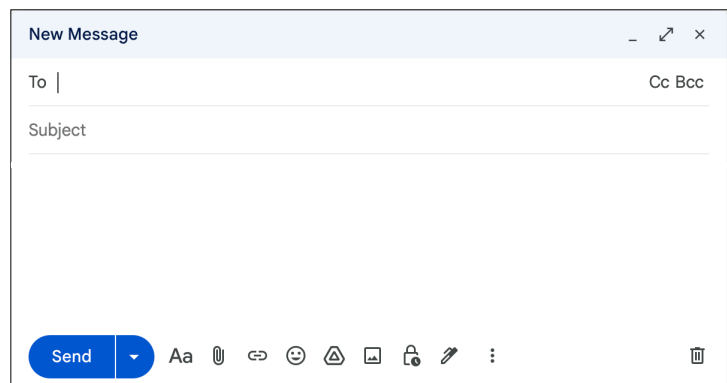
Step 2: Type `www.gmail.com` in the Address Bar and press the Enter key.

Step 3: Click the Sign in button.

Step 4: Enter your e-mail address or phone number and click Next.

Step 5: Enter your password and click Next.

Step 6: Your Gmail Inbox opens, and you can view your e-mails.



Sending an E-Mail

Once you have signed in, you can send an e-mail to anyone who has an e-mail address.

Steps to Send an E-mail

Step 1: Click the Compose button.

Step 2: The New Message window opens.

Step 3: Fill in the required fields.

To: Enter the recipient's e-mail address.

Cc (Carbon Copy): Sends a copy of the e-mail to another person.

Bcc (Blind Carbon Copy): Sends a copy without showing the recipient's address to others.

Subject: Type a short title for the e-mail.

Message: Type the main content of your e-mail.

Step 4: If required, click the Attach files button to attach a document, picture, or other file.

Step 5: Click the Send button to send the e-mail.

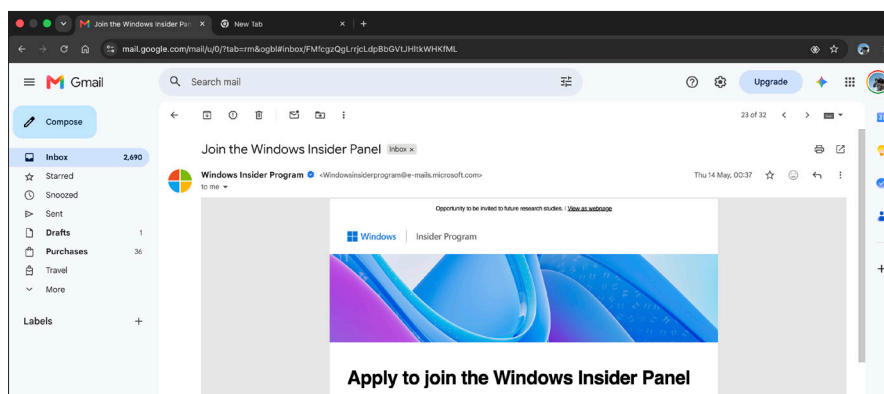
Reading an E-Mail

When someone sends you an e-mail, it is received in your Inbox. The Inbox displays all the e-mails you have received. Unread e-mails usually appear in bold, making them easy to identify. You can open and read any e-mail by clicking on its subject or the sender's name.

If an e-mail contains an attachment, an attachment icon appears beside it. You can open or download the attached file if required.

Steps to Read an E-mail

Step 1: Sign in to your Gmail account.



Step 2: Click the Inbox folder.

Step 3: Click the Subject or Sender's Name of the e-mail you want to read.

Step 4: The selected e-mail opens on the screen.

Step 5: If the e-mail has an attachment, click it to view or download the file.

Signing Out

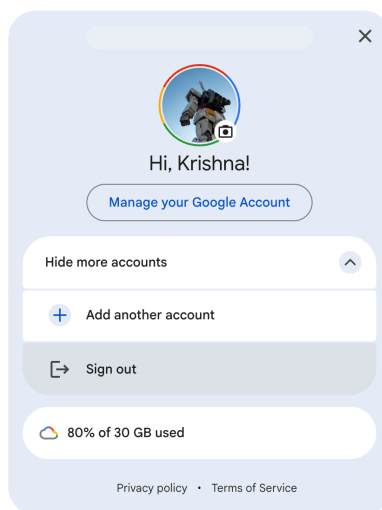
It is a good habit to sign out of your e-mail account after completing your work, especially when using a shared or public computer. Signing out helps keep your account safe and prevents others from accessing your personal e-mails.

Steps to Sign Out

Step 1: Click on your Profile icon at the top-right corner of the Gmail window.

Step 2: Click the Sign out button.

Your Gmail account is now safely closed.



Activity

With the help of your teacher, compose an e-mail to your friend inviting them to your birthday party. Fill in the details below.

To: _____

Subject: _____

Message:

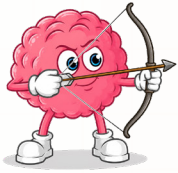




Quick Recap

- *The Internet is a global network of millions of computers connected worldwide.*
- *A computer system, modem, web browser, and ISP are required to access the Internet.*
- *Dial-up, Broadband, Wi-Fi, Mobile Internet, and Hotspots are common Internet connections.*
- *A URL is the unique address of a website.*
- *A web browser helps users open websites and search for information.*
- *E-mail is a fast way of sending and receiving electronic messages.*
- *Every e-mail address consists of a username, @ symbol, and domain name.*
- *Gmail allows users to create, send, receive, and manage e-mails.*





TEST YOUR SKILLS



A. Choose the correct option

1. Which of the following is used to connect a computer to the Internet?
a. Keyboard ☐ b. Printer ☐
c. Modem ☐ d. Scanner ☐
2. Which of the following is a web browser?
a. Gmail ☐ b. Google Chrome ☐
c. MS Word ☐ d. Paint ☐
3. Which Internet connection uses radio waves to provide Internet access?
a. Dial-Up Connection ☐ b. Broadband Connection ☐
c. Wi-Fi ☐ d. Telephone Line ☐
4. Which symbol is an essential part of every e-mail address?
a. # ☐ b. & ☐
c. % ☐ d. @ ☐
5. Which button is used to create a new e-mail?
a. Compose ☐ b. Send ☐
c. Reply ☐ d. Forward ☐

B. Fill in the Blanks

1. A website is opened by typing its _____.
2. An _____ provides Internet services to users.
3. The _____ connects millions of computers worldwide.
4. The _____ tells what an e-mail is about.
5. New e-mails are received in the _____.



C. State True or False

1. Every website has a unique URL. _____
2. Wi-Fi requires cables to connect devices. _____
3. An e-mail can include attachments. _____
4. Gmail is an Internet Service Provider. _____
5. You should sign out after using your e-mail on a public computer. _____

D. Answer The Following Questions

1. What is a Web Browser?
2. Name any four requirements needed to connect to the Internet.
3. What is an e-mail address?
4. Write the steps to open a website using a URL.
5. Write the steps to sign out from your Gmail account.

E. Competency-Based Questions

- a. Riya wants to search information about the Solar System on the Internet. Which application should she use, and what should she type to open the website?
- b. Aman wants to send his science project to his teacher by e-mail. Which field should he fill with the teacher's e-mail address, and which button should he click to send the e-mail?



Lets Solve

1. Neha has prepared a presentation for her class. She wants to send it to her friend through Gmail. Explain the steps she should follow.
2. Rahul forgot the password of his Gmail account. Which feature can help him recover his account, and why is it important to keep a recovery mobile number or e-mail address?
3. Meera has created a Gmail account. She wants to send the same invitation e-mail to her five classmates without showing each student's e-mail address to the others. Which e-mail field should she use?





Practical Exercises

Write an E-mail to your principal asking for 1 week holiday.

Lets Explore:

Visit an educational website with the help of your teacher or parents. Write the name of the website and mention three useful things you learned from it.

Lets Get Better:

Discuss with your classmates the safe practices to follow while using the Internet and e-mail. Prepare a list of five Internet safety rules.

Teacher's Note

- Demonstrate how to open a web browser and visit an educational website using a URL.
- Help students compose, send, and read a sample e-mail using a practice account.
- Explain the importance of strong passwords and safe Internet practices.



Data Processing



Let's Warm Up the Brain

Learning Goals

- Data & Information
- Sorting Data
- Representing Information
- Decoding

Your mother's birthday is coming soon. Help her by preparing a list of the items to be purchased for the celebration.

S. No.	Items Required	Quantity	Price

In the previous class, we learnt how to use different visualisation tools to organise and display data. However, simply collecting data is not enough. It must be arranged, processed and presented properly so that it becomes meaningful and easy to understand. This process helps us make better decisions and find the required information quickly.

In this chapter, we will learn about data and information, sorting data, representing information, and decoding messages. These concepts help us understand how raw facts are converted into useful information.

Data and Information

Data refers to a collection of raw facts and figures. It may consist of numbers, words, symbols or observations that have not yet been processed. By itself, data may not have any clear meaning. When data is organised, arranged and processed, it becomes information. Information is meaningful and helps us understand a situation or make decisions.

For example, the marks obtained by students in an examination are data. When these marks are arranged in a table and analysed to identify the highest scorer or calculate the average marks, they become information.

Name	Marks Obtained
John	95
Thomas	87
Arthur	64



Sorting Data

Sometimes, data is collected in a random order, making it difficult to understand or find the required information. Sorting is the process of arranging data in a particular order, such as alphabetical order, numerical order, ascending order (smallest to largest) or descending order (largest to smallest). Sorting helps us organise data neatly. It saves time, reduces confusion and makes it easier to compare similar items. Computers use sorting to quickly search and manage large amounts of data.



Name of the Items	Number of Items
1. Shuttle Cock	3
2. Football	4
3. Cricket Ball	6
4. Cricket Bat	2
5. Volley Ball	2
6. Basket Ball	2

Now, let's sort the number of items in descending order. You will get the following data.

Name of the Items	Number of Items
1. Cricket Ball	6
2. Football	4
3. Shuttle Cock	3
4. Cricket Bat	2
5. Volley Ball	2
6. Basket Ball	2



Representing Information

After data has been processed, it should be presented in a simple and meaningful way. Representing information means displaying information in a clear format so that it can be understood easily. Information can be represented in different ways depending on the purpose. Some common methods are tables, pictures, maps and pictograms.

Table

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
8:00 – 8:45	English	Mathematics	Science	Hindi	Mathematics	English
8:45 – 9:30	Mathematics	English	Hindi	Science	Science	Mathematics
9:30 – 10:15	Science	Hindi	Mathematics	English	Hindi	Science
10:15 – 10:30 SHORT BREAK						
10:30 – 11:15	Hindi	Science	English	Mathematics	English	Hindi
11:15 – 12:00	Social Science	Computer	Social Science	Computer	Social Science	Computer
12:00 – 12:45	Computer	Social Science	Computer	Social Science	Computer	Social Science
12:45 – 1:30 LUNCH BREAK						
1:30 – 2:15	Physical Education	Art & Craft	Library	Physical Education	Art & Craft	Library
2:15 – 3:00	Value Education	Music	Value Education	Music	Value Education	Art & Craft

Pictures

Maps

Pictograms

Decoding



Could you understand what Mr Grumpy wants to say? To understand what he said, we have to decode the message.

Sometimes, messages are written using special symbols, letters or numbers instead of normal words. Such messages are called coded messages. To understand these messages, we need to convert the code into a language that we can read.

Decoding is the process of converting a coded message into understandable language. It helps us find the original meaning of a secret or coded message.

A 1	B 2	C 3	D 4	E 5	F 6	G 7
H 8	I 9	J 10	K 11	L 12	M 13	N 14
O 15	P 16	Q 17	R 18	S 19	T 20	U 21
V 22	W 23	X 24	Y 25	Z 26		

So Grumpy's message is decoded as:

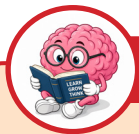
H 8	E 5	L 12	L 12	O 15
--------	--------	---------	---------	---------

Hence, the secret message after decoding is "Hello".



Fact

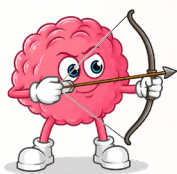
Ants build efficient networks for finding food and communicating. Scientists have discovered that the way ants organise their paths has inspired computer scientists to design better and more efficient communication and transport networks.



Quick Recap

- Data refers to raw facts and figures.
- Information is processed data that is meaningful and useful.
- Sorting means arranging data in a specific order to make it easier to understand.
- Information can be represented using tables, pictures, maps and pictograms.
- Decoding is the process of converting a coded message into understandable language.





TEST YOUR SKILLS



A. Choose the correct option

- Which of the following is an example of information?
a. Random numbers () b. Processed Exam Result ()
c. Unarranged data () d. Secret code ()
- Which method uses pictures or symbols to represent information?
a. Tables () b. Maps ()
c. Pictograms () d. Charts ()
- The process of converting a coded message into understandable language is called?
a. Encoding () b. Processing ()
c. Sorting () d. Decoding ()
- Which type of sorting arranges data from the smallest value to the largest?
a. Alphabetical Order () b. Descending Order ()
c. Ascending Order () d. Random Order ()
- Which of the following is the best way to represent a school timetable?
a. Table () b. Maps ()
c. Pictogram () d. Picture ()

B. Answer The Following Questions

- What is information?
- Why is sorting data important?
- Name any four ways of representing information.
- What is decoding? Give one real-life example.

C. Competency-Based Questions

- The school library has received 15 Science books, 10 English books, 18 Computer books and 12 Mathematics books. Arrange the books in ascending order according to their quantity. Which subject has the maximum number of books?
- Aman receives the coded message "DPNQVUFS". Each letter has been shifted one step forward in the alphabet (A – B, B – C, etc.). Decode the message and write the original word.





Activity

Identify and write the names of the following signs.























Explore

Observe your surroundings and collect information about five different vehicles passing your school in 15 minutes.

Represent the collected information in a table.

Vehicle	Number Observed
Car	
Bus	
Motorcycle	
Bicycle	
Truck	

After completing the table, answer:

Which vehicle was seen the most and which vehicle was seen the least?





Activity

Create a colourful poster using Paint on the topic: “Digital Safety Begins with You”

Include at least five safety rules such as:

- Create strong passwords.
- Never share personal information.
- Use trusted websites.
- Install antivirus software.
- Think before clicking unknown links.

Add suitable pictures, symbols and colours to make your poster attractive.

Teacher's Note

Explain the importance of algorithms, flowcharts, and binary representation in solving problems. Encourage students to write simple IF-THEN-ELSE conditions using real-life situations before introducing programming languages.



Conditional Blocks in Scratch

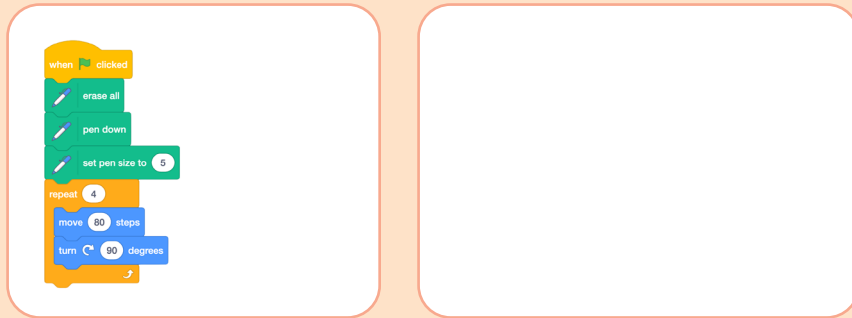


Let's Warm Up the Brain

Learning Goals

- Identify Blocks Shape
- Sensing Blocks
- Variables
- Use of Conditional Control Blocks
- Use of Loops Blocks
- Creating a Game

Check out the given code and draw the output.



Blocks Shapes in Scratch

Scratch programs are created by joining different blocks together. Each block has a special shape that tells us how it can be used in a script. Some blocks start a script, some perform actions, some check conditions, while others store or report information.

Scratch blocks are divided into six different shapes: **Hat Blocks**, **Stack Blocks**, **Boolean Blocks**, **Reporter Blocks**, **C Blocks**, and **Cap Blocks**.

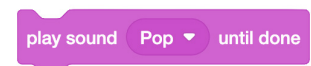
Hat Blocks

Hat Blocks are used to begin a script. They are called Hat Blocks because their curved top looks like a hat. Every script starts with a Hat Block, so no other block can be placed above it. These blocks tell the sprite when to begin performing an action, such as when the Green Flag is clicked, a key is pressed, or the sprite is clicked.



Stack Blocks

Stack Blocks are the most commonly used blocks in Scratch. They contain the commands that tell a sprite what to do, such as moving, speaking, changing costumes, or playing sounds. These blocks have a notch at the top and a bump at the bottom, allowing them to fit together like puzzle pieces to form a script.



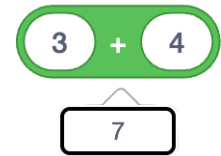
Boolean Blocks

Boolean Blocks are used to check whether a condition is true or false. They have pointed ends and return only two possible values – True or False.



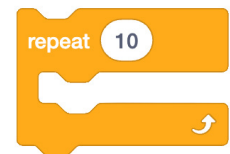
Reporter Blocks

A Web Browser is application software used to open and view websites. It allows users to search for information and access web pages on the Internet. Some popular web browsers are Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.



C Blocks

C Blocks are named after their shape, which looks like the letter C. They are also called Wrap Blocks because they hold other blocks inside them. These blocks are mainly used for repeating actions or checking conditions while the blocks placed inside them are executed.



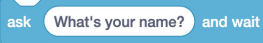
Cap Blocks

Cap Blocks are used to end a script. They have a flat bottom, showing that no more blocks can be attached below them. These blocks stop the execution of a script and are usually placed at the end whenever required.



Sensing Blocks

Sensing Blocks help a sprite gather information from the user or the stage while a program is running. They allow the sprite to detect actions such as a key being pressed, the mouse pointer touching a sprite, or a colour being detected.

-  ask [] and wait block displays a question on the stage and waits for the user to type an answer before the script continues.
-  touching [] ? block checks whether the sprite is touching another sprite, the mouse pointer, or the edge of the stage. It returns True if the condition is met; otherwise, it returns False.
-  touching colour [] ? block checks whether the sprite is touching the selected colour. It is useful for creating games and interactive animations.
-  colour [] is touching [] ? block checks whether one selected colour is touching another selected colour. It is often used to detect collisions between coloured objects.



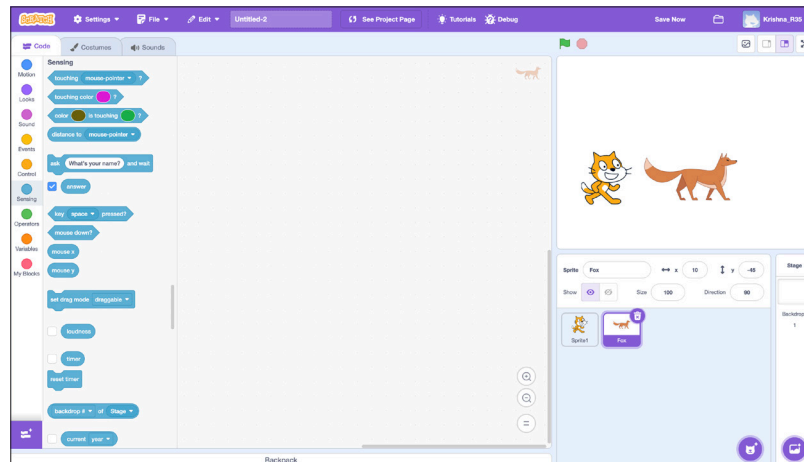
- **key [] pressed?** block checks whether the selected key on the keyboard is being pressed. It helps control a sprite using the keyboard.

Sensing an Input

Let us create a simple script to understand how a sprite can receive input from the user.

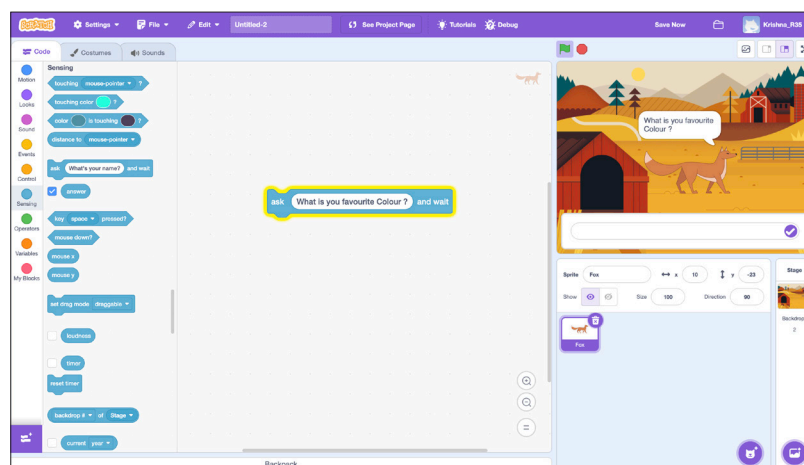
Step 1: Click on the Sensing block category from the Blocks Palette.

Step 2: Choose a sprite of your choice and add a suitable backdrop to the stage.



Step 3: Drag the ask [] and wait block into the Script Area. Type a question such as “What is your favourite colour?” inside the block.

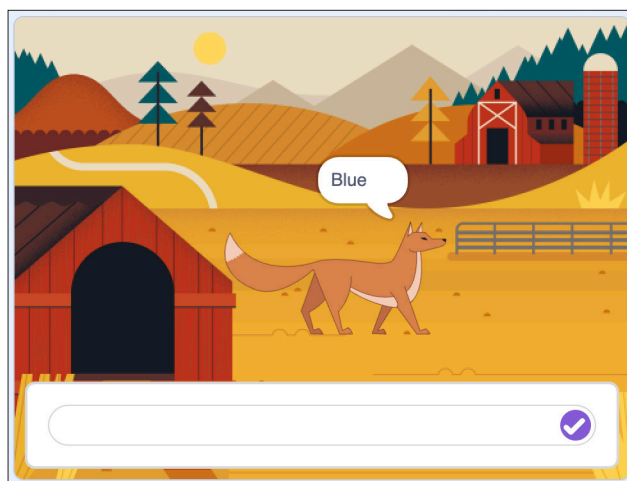
Step 4: Click the block to run it. A question appears in a speech bubble, and an input box is displayed on the stage. Type your answer and press Enter.



Step 5: To display the typed answer on the stage, use the answer reporter block. You can also place the answer block inside a say block to make the sprite speak the user’s response for a few seconds.



The sprite now asks the question, waits for the user's reply, and uses the entered answer in the program. This makes the Scratch project interactive and enjoyable.



Let's Catch Up

Answer the following questions:

1. Which block is used to ask a question and wait for the user's reply?
2. Which Scratch block checks whether a sprite is touching another sprite or the mouse pointer?
3. Which block is used to detect whether a particular key on the keyboard is pressed?

Variables

A variable is a named container used to store information in a Scratch project. The value stored in a variable can change while the program is running, which is why it is called a variable. Variables help keep track of information such as scores, lives, levels, time, or the number of collected objects in a game. Whenever the value of a variable changes, Scratch automatically updates it during the execution of the program. This makes variables very useful for creating games, quizzes, animations, and other interactive projects. Variables can store numbers as well as text, depending on the purpose of the program.

For example, in a racing game, a variable named Score can increase whenever the player collects a coin. Similarly, a variable named Player Name can store the name entered by the user at the beginning of the game.



Types of Variables

Scratch allows you to create two types of variables based on who can use them.

Numeric Variable: A variable that stores numbers is called a Numeric Variable. They can store whole numbers as well as decimal numbers. For example, 25, 100, -15, and 45.5 are numeric values.

String Variable: A variable that stores text is called a String Variable. It can hold letters, words, sentences, or a combination of letters and numbers. For example, "Riya", "Welcome", "Blue", and "Class 5" are string values.

Creating a Variable in Scratch

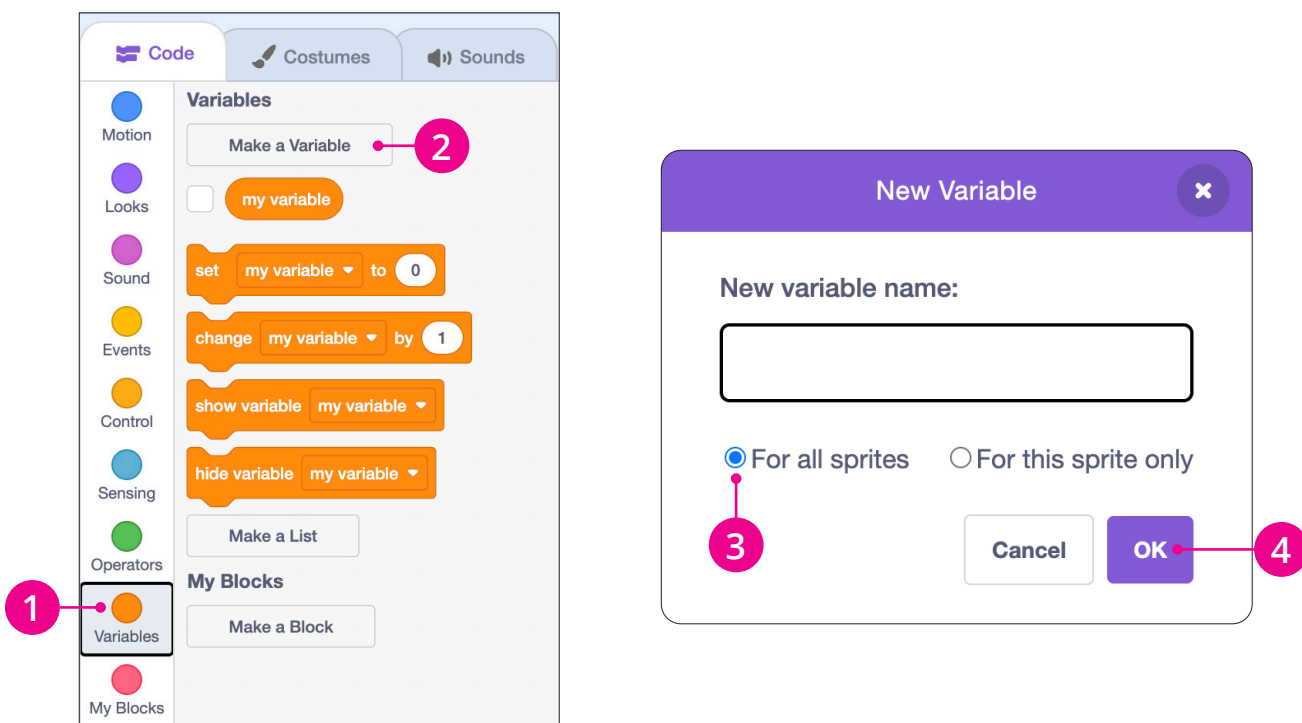
To create a variable in scratch follow the steps given below:

Step 1: Click on the Variables category in the Blocks Palette.

Step 2: Click the Make a Variable button. The New Variable dialog box appears. Type a suitable name for the variable, such as Score, Points, or Lives.

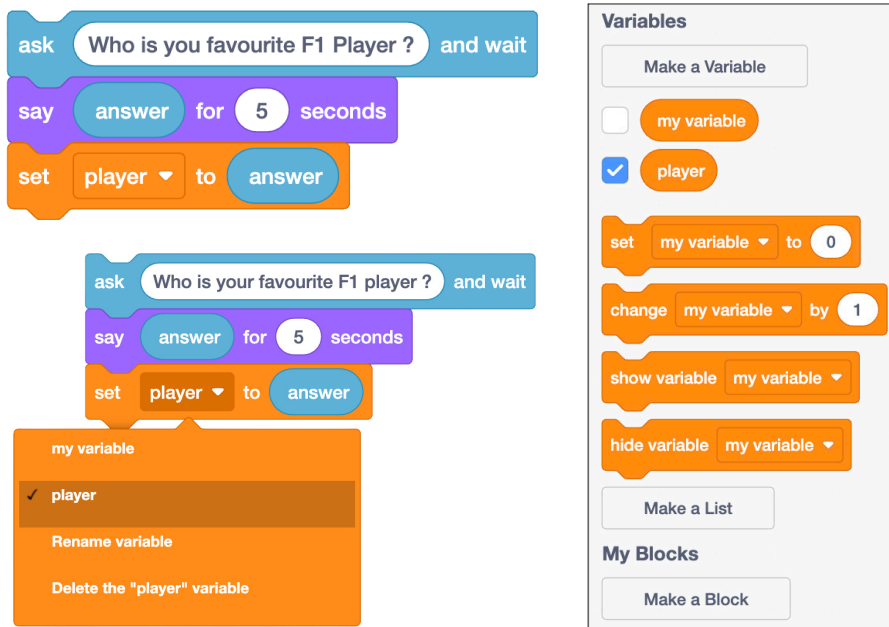
Step 3: Choose whether you want the variable to be available For all sprites or For this sprite only. Select For all sprites if you want this variable to appear for all the sprites otherwise select For this sprite only if you want this variable to appear for the selected sprite only.

Step 4: Click the OK button. The new variable is created and added to the Variables category.

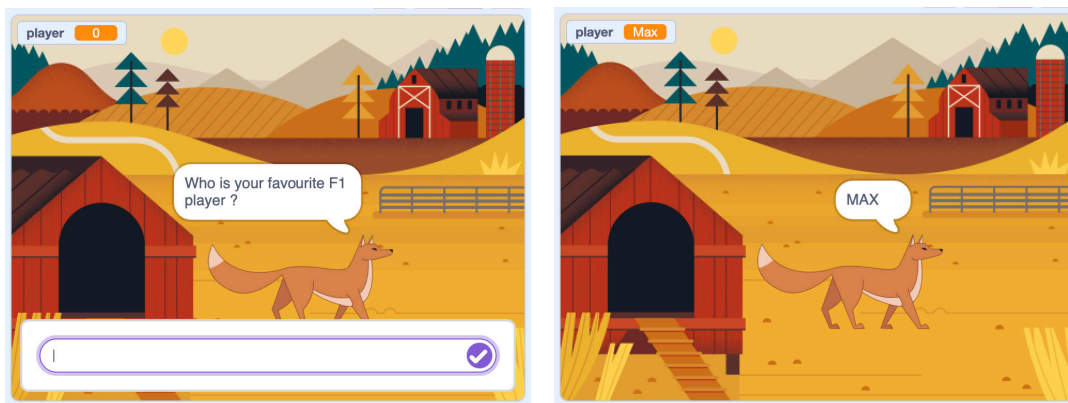


The variable appears in the list of variable blocks.

Now create the following script:



The sprite will ask you for your favourite F1 player. Enter your favourite player and press Enter. The name of your favourite player will appear in the speech bubble of your sprite. The name of the player then appears in the answer as well as the variable name on the stage.



Use of Conditional Control Blocks

In our daily life, we make decisions based on different situations. For example, before carrying an umbrella, we check the weather. If it is raining, we carry an umbrella. Otherwise, we leave it at home. Similarly, in Scratch, a program can perform different actions depending on whether a condition is true or false.

Suppose you are creating a game in Scratch in which a player collects coins. There can be two situations:

- a. If the player collects a coin, the score should increase by one.
- b. If the player does not collect a coin, the score should remain the same.

This can be written as:

```
1
2 if
3   sprite is touching the coin
4 then
5   increase score by 1
```

```
1 If
2   sprite is touching the coin
3 Then
4   increase score by 1
5 Else
6   keep the score unchanged
```

To perform such tasks, Scratch provides Conditional Control Blocks. These blocks check whether a given condition is true or false and execute the appropriate instructions. The two conditional blocks available in Scratch are:

If...Then Block: These blocks check a condition. If the condition is true, the blocks placed inside it are executed. If the condition is false, the blocks inside the block are skipped, and the program continues with the remaining script.

For example, suppose a sprite should say “Great Job!” only when the score is greater than 20. If the score is 20 or less, the sprite will not display any message.

If...Then...Else Block: This block is used when you want the program to perform one action if the condition is true and another action if the condition is false. This block contains two sections – Then and Else.

For example, suppose a quiz program checks the user’s answer. If the answer is correct, the sprite says “Correct Answer!” Otherwise, it says “Try Again!” Thus, one of the two actions is always performed depending on the result of the condition.



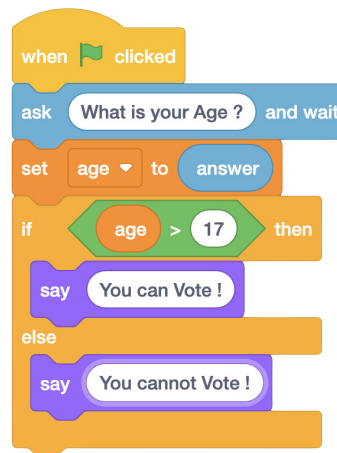
Tech Fact

Did You Know?

Conditional blocks help computers make decisions, just like humans do. They are used in many real-life applications such as traffic lights, automatic doors, elevators, and video games, where an action is performed only when a specific condition is true.



Suppose a person wants to cast a vote in an election. Before allowing the person to vote, the system checks the person's age using conditional control blocks.



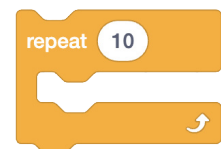
Use of Loop Blocks

There are many situations where we need to repeat the same task several times. For example, while practising handwriting, a student may write the same word ten times. Similarly, a traffic signal changes its lights continuously in the same order. Instead of writing the same instructions repeatedly, Scratch uses Loop Blocks to perform a set of instructions again and again.

Loop Blocks are available in the Control category. They help reduce the number of blocks in a script and make programs simpler, shorter, and easier to understand. Scratch provides different loop blocks for repeating actions in different ways.

Repeat Block

The Repeat block is used when an action has to be performed a fixed number of times. You can specify the number of repetitions, and the blocks placed inside it are executed exactly that many times. **For example:** if a sprite has to move 20 steps five times, the Repeat (5) block can be used. After completing five repetitions, the program automatically moves to the next block in the script.



Forever Block

The Forever block repeats the enclosed blocks continuously until the program is stopped manually. It is commonly used in games, animations, and interactive projects where an action needs to keep running throughout the program.

For example: a butterfly sprite can keep flying across the stage, or a fish can keep swimming in the water using the Forever block.



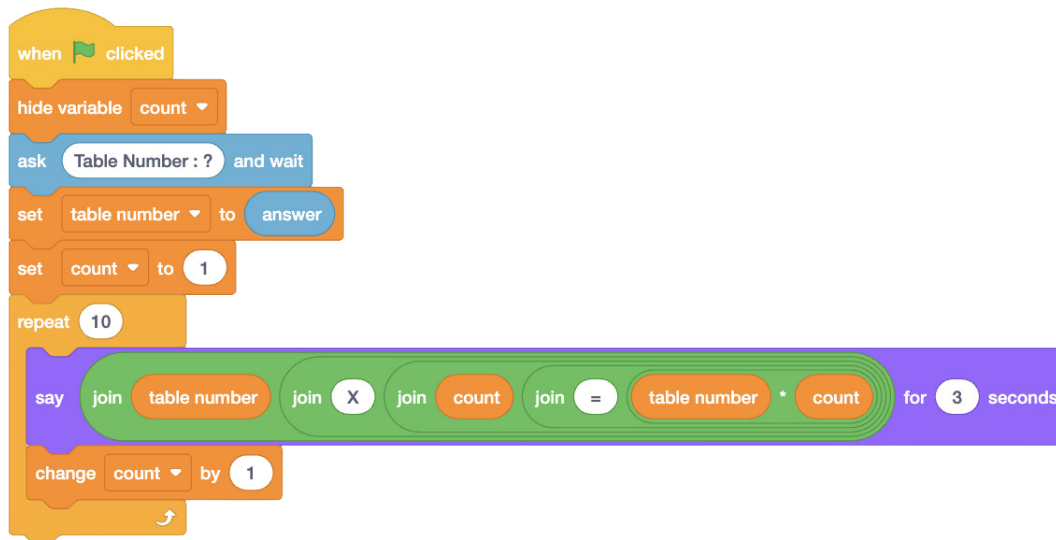
Repeat Until Block

The Repeat Until block repeats the enclosed instructions until a specified condition becomes true. As soon as the condition is satisfied, the loop stops and the program continues with the next instruction.



For example: in a maze game, a sprite can continue moving until it reaches the finish point. Once the sprite touches the finish line, the loop ends automatically.

Lets see the script to display the table of a number in Scratch.



The output of the program will be similar to this:

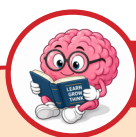
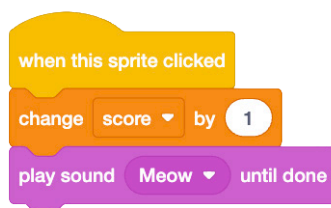


Creating a Game

Scratch provides an easy way to create fun and interactive games. A game is created by combining different Scratch blocks such as Events, Motion, Control, Sensing, Operators, and Variables. These blocks help control the movement of sprites, keep track of scores, detect user actions, and make decisions during the game. By using loops and conditional blocks, the game can respond automatically to different situations and become more interesting to play.

In this activity, we will create a simple game in which a player earns points by clicking on a moving sprite. The game begins when the Green Flag is clicked, and the score is set to 0. The sprite keeps moving to random positions on the Stage, making it challenging for the player to catch it. Whenever the player clicks on the sprite, the score increases by 1. The game continues until the player decides to stop it.

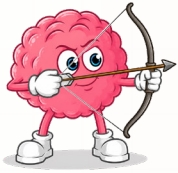
Follow the Script to make this game:



Quick Recap

- Scratch uses different block shapes, and each shape has a specific purpose.
- Sensing blocks help sprites interact with users and the environment.
- Variables store information that can change while a program is running.
- Conditional blocks help a program make decisions.
- Loop blocks repeat instructions automatically.
- Scripts are created by joining blocks together to perform tasks.
- Scratch can be used to create interactive games, animations, and stories.





TEST YOUR SKILLS



A. Choose the correct option

1. Which block is used to repeat a set of instructions continuously?
a. Repeat ☐ b. Forever ☐
c. Repeat Until ☐ d. If...Then ☐
2. Which of the following is a numeric variable?
a. 18 ☐ b. "Scratch" ☐
c. "Variable" ☐ d. "String" ☐
3. Which block is used to make decisions in Scratch?
a. Motion Block ☐ b. Sound Block ☐
c. Conditional Block ☐ d. Variable Block ☐
4. Which block is used to take input from the user?
a. Say Block ☐ b. Move Block ☐
c. Boolean Block ☐ d. Ask and Wait ☐

B. Fill in the Blanks

1. _____ blocks return either True or False.
2. _____ block repeats the instructions until the program is stopped.
3. Information that changes during a program is stored in _____.
4. The user's reply is stored in the _____ block.

C. State True or False

1. Reporter blocks can store numbers and text. _____
2. Hat blocks are placed at the end of a script. _____
3. The repeat until block stops when condition becomes true. _____
4. Variables can be used to store scores in a game. _____



D. Answer The Following Questions

1. What are Reporter blocks?
2. Why are variables used in Scratch?
3. What is the use of the Repeat Until block?
4. Name any two conditional control blocks used in Scratch.

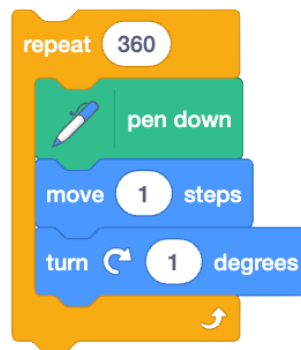
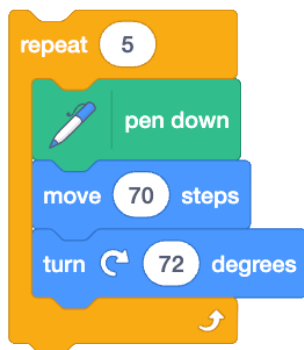
E. Competency-Based Questions

- a. Riya is creating a quiz in Scratch. She wants the sprite to display "Correct!" when the user enters the right answer and "Try Again!" when the answer is wrong. Which Scratch block should she use?
- b. Aarav wants his game character to keep moving until it touches the finish line. Which loop block is most suitable for this task and why?



Let's Solve

Write will the sprite do in the following cases:



Run these in scratch and find out.



F. Let's Explore

Explore any three Motion blocks in Scratch and find out how they affect the movement of a sprite. Share your observations with your classmates.

G. Let's Get Better

Always create your own Scratch projects and respect the work of others. Give credit when you use someone else's ideas or sprites, and encourage your friends to be honest and creative.

Teacher's Note

- *Demonstrate how variables can be used to keep score in a simple Scratch game.*
- *Encourage students to use conditional and loop blocks to create their own interactive projects.*



Concept of Smart Living



Let's Warm Up the Brain

Learning Goals

- Smart Homes
- Smart Home Devices
- Benefits of Smart Living
- Voice Assistants
- Smart Home Safety

Imagine you are a smart home designer. List any four smart features that you would like to include in your dream home.

1. _____
2. _____
3. _____
4. _____

Technology has become an important part of our everyday life. Today, many homes use smart devices that can perform tasks automatically and can be controlled through the Internet. These devices make our homes more comfortable, secure, and energy-efficient. From switching on lights to monitoring home security, smart technology helps us manage our daily activities with ease. Such homes are called Smart Homes, and they make living simpler, safer, and more convenient.

Smart Homes

A smart home is a house where different devices are connected through the Internet and can be controlled automatically or from a smartphone. These devices make everyday work easier, improve comfort, and help save time and energy. Smart homes also increase safety by allowing people to monitor and manage their homes from anywhere.

Benefits of Smart Homes

The following are the benefits of Smart Homes:

Save Energy: Smart homes help save electricity by automatically switching off lights, fans, and other appliances when they are not in use. This reduces unnecessary power consumption.

Better Security: Smart security devices such as cameras, smart locks, and video doorbells help keep the home safe. They can send alerts to the owner's smartphone if any unusual activity is detected.

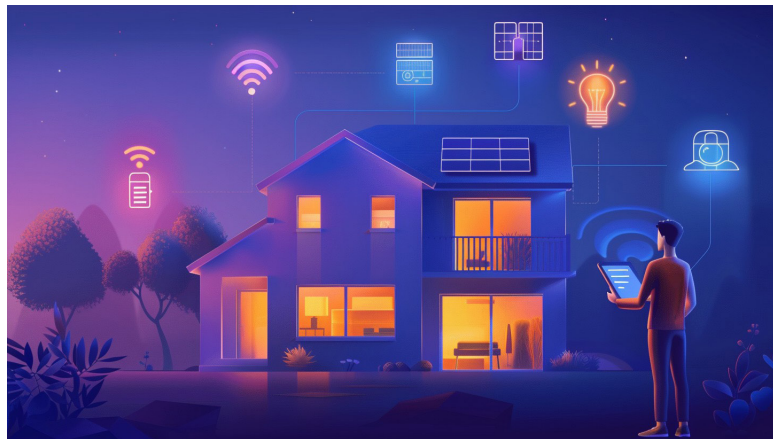


Remote Control: Smart home devices can be controlled from anywhere using a smartphone or tablet. For example, users can switch on lights, adjust the air conditioner, or check security cameras even when they are away from home.

Greater Comfort: Smart devices can be programmed to work according to the user's daily routine. They can automatically adjust lighting, room temperature, and other settings to provide a comfortable living environment.

Saves Time and Effort: Many everyday tasks, such as turning on lights, locking doors, or operating appliances, can be done automatically or with simple voice commands. This saves time and makes life easier.

Quick Emergency Alerts: Smart devices such as smoke detectors and water leak sensors can immediately warn users about emergencies. This allows quick action and helps prevent major damage.



Devices used in Smart Homes

Modern smart homes use Internet-connected devices that can be controlled remotely. Many of these devices also use Artificial Intelligence (AI) to understand user preferences and perform tasks automatically. Some common smart home devices are described below.



Tech Fact

Did you know? Some modern smart homes use Artificial Intelligence (AI) to learn the daily habits of the people living in them. For example, the system can remember when you usually wake up, switch on the lights before you enter a room, or adjust the room temperature automatically.



Smart TV

A Smart TV connects to the Internet and offers much more than watching television channels. It allows users to stream videos, listen to music, browse online content, and install useful applications. Many Smart TVs can also be controlled using voice commands.



Video Doorbells

A video doorbell lets homeowners see and speak to visitors before opening the door. It has a built-in camera, microphone, and speaker that work through a smartphone. This improves home security and allows users to answer the door even when they are away.



Smart Cameras

Smart cameras keep an eye on activities inside and outside the home. They record videos, detect movement, and send alerts whenever unusual activity is noticed. These cameras help homeowners monitor their property from anywhere.



Smart Smoke Detectors

A smart smoke detector senses smoke and gives an alarm when there is a risk of fire. It can also send notifications to the owner's smartphone, allowing quick action even if no one is at home.



Smart Lighting

Smart lighting systems allow users to switch lights on or off using a smartphone or voice commands. They can adjust brightness, set timers, and schedule lighting according to daily routines. This helps save electricity and creates a comfortable environment.



Smart Speakers

Smart lighting systems allow users to switch lights on or off using a smartphone or voice commands. They can adjust brightness, set timers, and schedule lighting according to daily routines. This helps save electricity and creates a comfortable environment.



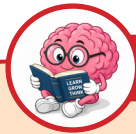


Activity

Imagine you have Rs.1,00,000 to build a smart home. Choose any five smart devices from the list below and explain why you selected them.

- () Smart TV
- () Smart Speaker
- () Smart Camera
- () Smart Bulb
- () Smart Lock
- () Video Doorbell
- () Smart Smoke Detector
- () Smart Thermostat

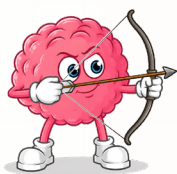
Why did you choose these devices?



Quick Recap

- Smart homes use Internet-connected devices to make everyday life easier.
- Smart devices can be controlled remotely using a smartphone or voice commands.
- Smart homes improve comfort, security, and energy efficiency.
- Popular smart home devices include Smart TVs, video doorbells, smart cameras, smart smoke detectors, smart lights, and smart speakers.





TEST YOUR SKILLS



A. Choose the correct option

- Which smart device helps you talk to visitors before opening the door?
a. Smart TV () b. Video Doorbell ()
c. Smart Speaker () d. Smart Camera ()
- Which smart device helps save electricity by controlling lights?
a. Smart Lighting () b. Smart Speaker ()
c. Smart Camera () d. Smart TV ()
- Which device can answer questions / play music using voice commands?
a. Smart TV () b. Smoke Detector ()
c. Smart Speaker () d. Smart Camera ()
- Which of the following is used to monitor activities around the house?
a. Smart TV () b. Smart Lighting ()
c. Video Doorbell () d. Smart Camera ()

B. Fill in the Blanks

- Smart homes help in saving _____.
- A smart _____ accepts voice commands.
- Smart homes use the _____ to connect devices.
- A smart _____ detector warns us about fire.

C. State True or False

- Smart homes can be controlled using a smartphone. _____
- Smart speakers cannot play music. _____
- Smart cameras help improve home security. _____
- Smart lighting helps reduce electricity consumption. _____
- Smart smoke detectors warn people when smoke is detected. _____



D. Answer The Following Questions

1. What is a Smart Home?
2. Write any four benefits of Smart Homes.
3. Name any four smart devices used in Smart Homes.
4. How does a Smart Speaker make our daily life easier?

E. Competency-Based Questions

- a. You are leaving your home for a family trip. Which smart devices would you use to keep your home safe? Explain your answer.
- b. Your family wants to save electricity at home. Which smart home devices would you suggest? How will they help?



Lets Solve

Rearrange the letters and write the names of the smart devices.

- a. MTRAS GITNLIHG _____
- b. TRMAS REPEAKS _____
- c. ODEIV OORDBELL _____



Lets Explore

Visit a nearby electronics store or search on the Internet to find any four smart home devices. Write one feature of each device.

Smart Device	One Feature
1. _____	_____
2. _____	_____
3. _____	_____
4. _____	_____



F. Let's Get Better

Smart technology is making our homes safer and more comfortable.

Think and write: How can smart home devices help elderly people or persons with disabilities live more safely and independently?

Teacher's Note

- *Explain how smart home devices improve safety, comfort, and energy efficiency.*
- *Encourage students to identify smart devices used in their homes or neighbourhood and discuss their practical uses.*

